

General Relativity I

Lecture Notes

Manifolds, tangent spaces, differential forms,
connections, and Lorentzian geometry

Metric signature $(+, -, -, -)$

Smooth means C^∞

Contents

1	From affine spacetime to local geometry	1
1.1	Linear structure at a point	1
2	Smooth manifolds and local coordinates	3
2.1	Charts and smooth structures	3
2.2	Smooth maps	4
2.3	Local coordinates and global structure	5
3	Curves, tangent vectors, and derivations	6
3.1	Differentiating along a curve	6
3.2	Local functions and tangent vectors	7
3.3	The coordinate basis	8
3.4	Recovering the velocity of a curve	9
4	Differentials and coordinate transformations	10
4.1	The differential of a map	10
4.2	Changing coordinates at a fixed point	11
5	Cotangent spaces, bundles, and fields	12
5.1	Covectors and the differential of a function	12
5.2	Assembling the tangent spaces	13
5.3	Sections and tensor fields	14
6	Tangent geometry of the two-sphere	15
6.1	Intrinsic vectors and the ambient tangent plane	15
6.2	Spherical coordinates and coordinate vectors	16
6.3	Comparing vectors at different points	17
7	Connections and parallel transport	17
7.1	Covariant differentiation	17
7.2	Differentiation along a curve	19
8	Differential forms and exterior algebra	21
8.1	The wedge product	21
8.2	Components and normalization	22
9	Exterior differentiation and pullback	23
9.1	The exterior derivative	23

9.2	Pullback of forms	24
9.3	Local and global potentials	26
10	Vector fields, flows, and Lie derivatives	27
10.1	The Lie bracket	27
10.2	Local flows	28
10.3	Lie differentiation	29
10.4	Lie derivatives of differential forms	30
10.5	Exterior differentiation in a general frame	31
11	Metrics and measurement	32
11.1	Metrics and signature	32
11.2	Metric duality and the inverse metric	33
12	Sphere and spacetime metrics	34
12.1	The round metric on the sphere	34
12.2	Lorentzian line elements and proper time	35
12.3	Local Lorentz frames	36
12.4	Metric invariance under a flow	37
12.5	From the metric to covariant geometry	37
A	The Levi–Civita connection	38
A.1	Covectors and tensor fields	38
A.2	Torsion and metric compatibility	39
A.3	Existence and uniqueness	39
A.4	Parallel transport and scalar products	41
A.5	The sphere connection and transport around a latitude	42
A.6	Coordinates inertial at one event	43
B	Boost coordinates in Minkowski spacetime	44
B.1	The chart and its metric	44
B.2	Boost symmetry and covariant differentiation	44
B.3	An accelerated clock	45

1 From affine spacetime to local geometry

SOURCE SLIDE 1

In special relativity, spacetime is an affine space equipped with the Minkowski metric. Given two events p and q , their difference $q - p$ is a well-defined displacement vector. All such displacements belong to the same vector space V . A vector $v \in V$ can describe a displacement starting at any event, taking p to $p + v$ or q to $q + v$. The arrows from p to $p + v$ and from q to $q + v$ therefore represent the same displacement vector, even though they start at different events.

In general relativity, spacetime is described by a smooth four-dimensional manifold M equipped with a Lorentzian metric g . No global affine structure is assumed. Each event p has a tangent space $T_p M$. A smooth parametrized curve passing through p determines a tangent vector there, given by its derivative. The space $T_p M$ consists of all vectors obtained in this way. Vectors in $T_p M$ can be added and subtracted, but a vector at another event q belongs to a different space, $T_q M$. The manifold structure alone does not tell us which vector in $T_q M$ should correspond to a given vector in $T_p M$. A connection provides a rule for making this comparison by transporting vectors along a chosen curve between the two events.

This generalization must be distinguished from a change of coordinates. Changing coordinates assigns different numerical labels to the same events, without changing the geometry. For example, a nonlinear coordinate change can make the components of the Minkowski metric depend on position. Returning to inertial Cartesian coordinates makes them constant again. The position dependence in the other coordinates does not indicate curvature.

General relativity allows both flat and curved spacetimes. If curvature is nonzero at an event p , no coordinates can make all metric components constant throughout a neighborhood of p . One can still arrange $g_{\mu\nu}(p) = \eta_{\mu\nu}$ at the single event, where $\eta_{\mu\nu}$ is the constant Minkowski matrix. This does not make $g_{\mu\nu}(x) = \eta_{\mu\nu}$ in a neighborhood. The relevant distinction is therefore not whether the displayed components vary, but whether the metric can be made constant throughout a coordinate neighborhood.

We begin by making local coordinates and tangent vectors precise. This will let us describe fields and their derivatives before adding a metric. The familiar linear algebra of SR1 remains valid within each tangent space, while the new geometric structures explain how these spaces fit together.

1.1 Linear structure at a point

SOURCE SLIDES 2–4

The vector-space and tensor conventions of SR1 apply at each point of a manifold. Unless stated otherwise, scalars are real. Let V be an n -dimensional real vector space. Its dual space is

$$V^* = \text{Hom}_{\mathbb{R}}(V, \mathbb{R}),$$

where $\text{Hom}_{\mathbb{R}}(V, W)$ denotes the real vector space of linear maps from V to another real vector space W . Elements of V^* are covectors.

Let $e_1, \dots, e_n$ be a basis of V . Its dual basis $\varepsilon^1, \dots, \varepsilon^n$ satisfies $\varepsilon^i(e_j) = \delta^i_j$. Thus

$$v = v^i e_i, \quad \omega = \omega_i \varepsilon^i, \quad \omega(v) = \omega_i v^i.$$

An index repeated exactly twice in a term, once upstairs and once downstairs, is summed over its range. Latin coordinate indices run from 1 to n , and Greek spacetime indices run from 0 to 3. The pairing $\omega(v)$ requires no metric.

For nonnegative integers r, s , a tensor of type (r, s) belongs to

$$T_s^r(V) = V^{\otimes r} \otimes (V^*)^{\otimes s}.$$

A zeroth tensor power is $\mathbb{R}$, so a tensor of type $(0, 0)$ is a scalar. A tensor is a finite sum of tensor products. In the chosen bases it has a unique component expansion

$$T = T^{i_1 \dots i_r}_{j_1 \dots j_s} e_{i_1} \otimes \dots \otimes e_{i_r} \otimes \varepsilon^{j_1} \otimes \dots \otimes \varepsilon^{j_s}.$$

It acts multilinearly on r covectors and s vectors through their dual pairings. In particular, a bilinear form has type $(0, 2)$. These constructions do not require a metric.

A change of basis changes components, not the tensor. If $e'_a = A^i_a e_i$, then $\varepsilon'^a = (A^{-1})^a_i \varepsilon^i$. For a tensor of type $(1, 1)$, this gives

$$\mathbf{T}' = A^{-1} \mathbf{T} A, \quad \mathbf{T} = (T^i_j), \quad \mathbf{T}' = (T'^a_b).$$

This is the change-of-basis rule from SR1 for an arbitrary invertible matrix A .

Student work – basis-independent quantities

Using the change-of-basis law for a tensor of type $(1, 1)$, prove that the trace and determinant of its component matrix are independent of the chosen basis.

On a manifold, take $V = T_p M$ at each point p . Local bases and the matrices relating them may vary smoothly with p .

2 Smooth manifolds and local coordinates

SOURCE SLIDE 5

2.1 Charts and smooth structures

A coordinate system assigns n real numbers to each point in an open region of an n -dimensional manifold. These numbers label the point and need not be components of a position vector. A coordinate system need not cover the whole manifold, so its domain must be specified. Where two coordinate domains overlap, a transition map relates the two coordinate descriptions of the same points.

Definition 2.1: Topological manifold and coordinate chart

An n -dimensional *topological manifold* is a Hausdorff, second-countable topological space M in which every point lies in an open set U admitting a homeomorphism

$$\varphi : U \longrightarrow \varphi(U) \subset \mathbb{R}^n$$

onto an open subset of $\mathbb{R}^n$. The pair (U, φ) is a *coordinate chart*. Writing $\varphi(p) = (x^1(p), \dots, x^n(p))$ defines the coordinate functions $x^i : U \rightarrow \mathbb{R}$.

A homeomorphism is a continuous bijection with continuous inverse. The Hausdorff condition requires distinct points to have disjoint open neighborhoods. Second countability requires a countable collection of open sets such that every open set is a union of members of that collection. The existence of a Euclidean coordinate neighborhood around every point does not by itself guarantee that the whole space is Hausdorff or second countable. These two conditions are therefore imposed separately in the definition of a manifold.

Let (U, φ) and (V, ψ) be coordinate charts with $U \cap V \neq \emptyset$. Every point $p \in U \cap V$ has two coordinate descriptions,

$$x = \varphi(p), \quad y = \psi(p).$$

To obtain y from x , first apply φ^{-1} to recover the point p , then apply ψ to find its coordinates in the second chart. The composition therefore acts as

$$x = \varphi(p) \xrightarrow{\varphi^{-1}} p \xrightarrow{\psi} \psi(p) = y.$$

It changes the coordinates used to describe the point, not the point itself. The resulting *transition map* is

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \longrightarrow \psi(U \cap V).$$

The restriction to the overlap is necessary because the point must belong to the domains of both charts. Its inverse is $\varphi \circ \psi^{-1}$, which converts the second coordinate description back to the first.

Since $U \cap V$ is open and the charts are homeomorphisms onto open subsets of $\mathbb{R}^n$, both $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open in $\mathbb{R}^n$. The transition maps therefore act between open subsets of Euclidean space, so their smoothness can be tested using ordinary multivariable calculus.

Definition 2.2: Smooth atlas and smooth manifold

Two charts are *smoothly compatible* if the transition map $\psi \circ \varphi^{-1}$ is C^∞ on $\varphi(U \cap V)$ and its inverse is C^∞ on $\psi(U \cap V)$. Charts with disjoint domains are compatible by convention. A *smooth atlas* is a family of pairwise smoothly compatible charts whose domains cover M .

A smooth atlas is *maximal* if it contains every chart that is smoothly compatible with all of its charts. A *smooth structure* is a maximal smooth atlas. A topological manifold equipped with a smooth structure is a *smooth manifold*.

Any smooth atlas determines a unique maximal one by adjoining every chart compatible with all of its charts. To check compatibility of two added charts near a point, compose their transitions through a chart in the original atlas containing that point. Thus specifying a convenient covering atlas suffices to specify the smooth structure.

2.2 Smooth maps

Charts let us test smoothness using ordinary calculus. For a map between manifolds, choose coordinates on both the domain and the target, then test the resulting map between open subsets of Euclidean spaces.

Definition 2.3: Smooth map and diffeomorphism

Let M and N be smooth manifolds. A map $F : M \rightarrow N$ is *smooth* if, for every $p \in M$, there are charts (U, φ) and (V, ψ) from their respective smooth structures, with $p \in U$ and $F(U) \subset V$, such that

$$\psi \circ F \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

is smooth in the ordinary Euclidean sense.

A *diffeomorphism* is a smooth bijection whose inverse is smooth.

This definition does not depend on the chosen charts. Replacing either chart composes the coordinate representative with smooth transition maps on the relevant overlaps. In particular, a scalar field $f : M \rightarrow \mathbb{R}$ is smooth precisely when $f \circ \varphi^{-1}$ is smooth in every chart belonging to the chosen smooth structure. Their space is denoted by $C^\infty(M)$. Pointwise addition, scalar multiplication, and multiplication of functions make it a real algebra. This algebra will provide the functions on which tangent vectors act as differentiation rules.

Let (U, φ) and (V, ψ) be smooth charts on M . The coordinate representatives of a scalar field $f : M \rightarrow \mathbb{R}$ are

$$f_x = f \circ \varphi^{-1}, \quad f_y = f \circ \psi^{-1},$$

defined on $\varphi(U)$ and $\psi(V)$, respectively. For $p \in U \cap V$, write $x = \varphi(p)$ and $y = \psi(p)$. Then

$$f_y(y) = f_x(x) = f(p).$$

Equivalently, on $\psi(U \cap V)$,

$$f_y = f_x \circ (\varphi \circ \psi^{-1}).$$

The coordinate expressions may differ, but they assign the same value to the same point. This generalizes the scalar-field transformation law of SR1 from affine Lorentz coordinate changes to arbitrary smooth coordinate transitions.

2.3 Local coordinates and global structure

SOURCE SLIDE 6

The definition of a manifold guarantees a coordinate chart around every point. It does not guarantee a single chart whose domain is the whole manifold. Such a *global chart* would be a homeomorphism from M onto an open subset of $\mathbb{R}^n$.

Consider the unit sphere

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}.$$

It is compact because it is closed and bounded in $\mathbb{R}^3$. If a global chart $\varphi : S^2 \rightarrow W$ existed, its image W would be a nonempty open subset of $\mathbb{R}^2$. It would also be compact, since a homeomorphism preserves compactness. No nonempty open subset of $\mathbb{R}^2$ is compact, so such a chart cannot exist. We can instead cover the sphere with two charts, each omitting a different point.

Example 2.4: Stereographic coordinates on the sphere

Let $N = (0, 0, 1)$ and $S = (0, 0, -1)$ be the north and south poles. Define the open subsets

$$U_N = S^2 \setminus \{N\}, \quad U_S = S^2 \setminus \{S\}.$$

The subscript indicates the pole excluded from the chart domain.

For $p = (x, y, z) \in U_N$, stereographic projection from N assigns to p the intersection of the line through N and p with the plane $z = 0$. Identifying this plane with $\mathbb{R}^2$ gives a pair of coordinates. The line has the parametrization

$$N + t(p - N) = (tx, ty, 1 + t(z - 1)).$$

Its third coordinate vanishes when $t = 1/(1 - z)$. Applying the same construction from S gives the two maps $\varphi_N : U_N \rightarrow \mathbb{R}^2$ and $\varphi_S : U_S \rightarrow \mathbb{R}^2$, with

$$\varphi_N(x, y, z) = \left(\frac{x}{1 - z}, \frac{y}{1 - z} \right), \quad \varphi_S(x, y, z) = \left(\frac{x}{1 + z}, \frac{y}{1 + z} \right).$$

Each map assigns two coordinates to every point in its domain. Conversely, every pair $(u, v) \in \mathbb{R}^2$ determines exactly one point in the corresponding domain. Writing $q = u^2 + v^2$, the inverse maps are

$$\varphi_N^{-1}(u, v) = \frac{(2u, 2v, q - 1)}{1 + q}, \quad \varphi_S^{-1}(u, v) = \frac{(2u, 2v, 1 - q)}{1 + q}.$$

Both maps and their inverses are continuous, so they are homeomorphisms onto $\mathbb{R}^2$. In particular, $\varphi_N(S) = \varphi_S(N) = (0, 0)$. Thus the origin in the north-pole chart represents the south pole, while the origin in the south-pole chart represents the north pole.

The two chart domains cover the whole sphere, and their overlap excludes both poles,

$$U_N \cup U_S = S^2, \quad U_N \cap U_S = S^2 \setminus \{N, S\}.$$

In either chart, the coordinate image of this overlap is $\mathbb{R}^2 \setminus \{(0, 0)\}$. The transition map is therefore

$$\varphi_S \circ \varphi_N^{-1} : \mathbb{R}^2 \setminus \{(0, 0)\} \longrightarrow \mathbb{R}^2 \setminus \{(0, 0)\},$$

Example 2.4 (continued)

with

$$(\varphi_S \circ \varphi_N^{-1})(u, v) = \left(\frac{u}{u^2 + v^2}, \frac{v}{u^2 + v^2} \right).$$

The reverse transition $\varphi_N \circ \varphi_S^{-1}$ has the same formula. The denominator is nonzero throughout the stated domain, so both transitions are C^∞ . Consequently, (U_N, φ_N) and (U_S, φ_S) form a smooth atlas on S^2 .

The exclusion of $(0, 0)$ applies only to the transition map, not to either full chart image. The south pole is covered by U_N , and the north pole is covered by U_S . Neither pole belongs to the overlap, because neither has coordinates in both charts.

Student work – the transition between two sphere charts

Verify the inverse and transition formulas. Compute the Jacobian determinant of the transition map and show that it is nonzero everywhere on its domain.

SOURCE SLIDE 7

The smooth structure specifies which functions and curves on the manifold are smooth. Smooth transition maps ensure that this property does not depend on the chart used to test it. The smooth structure alone does not determine lengths, angles, causal cones, or parallel transport. A metric supplies scalar products in tangent spaces and, in the Lorentzian case, light cones. A connection supplies a rule for parallel transport along curves.

We used the representation of S^2 as a surface in $\mathbb{R}^3$ to construct its charts explicitly. The definition of a smooth manifold does not require such a surrounding space. The following constructions use smooth curves and functions on the manifold itself, without requiring an embedding in a larger Euclidean space.

Unless stated otherwise, all manifolds below are finite-dimensional, smooth, and without boundary. Smooth means C^∞ . In particular, all charts used for differentiation belong to the chosen smooth structure.

3 Curves, tangent vectors, and derivations

SOURCE SLIDE 8

3.1 Differentiating along a curve

A smooth parametrized curve is a smooth map $c : I \rightarrow M$, where $I \subset \mathbb{R}$ is an open interval. The parameter $t \in I$ specifies which point $c(t)$ on the curve is being considered. It need not represent physical time. Suppose $c(t_0) = p$. We want to define the tangent vector $\dot{c}(t_0)$, which describes how the curve passes through p .

In an affine space, this vector can be obtained from a difference quotient of points. On a general manifold, the subtraction $c(t_0 + h) - c(t_0)$ is not defined. Instead, let f be a smooth real-valued function near p . The composition $f \circ c$ is a real-valued function of one variable, so its derivative is defined by ordinary calculus. We set

$$\dot{c}(t_0)(f) := \left. \frac{d}{dt} \right|_{t=t_0} f(c(t)). \quad (3.1)$$

Thus $\dot{c}(t_0)$ is described by its action on smooth functions. Given f , it returns the rate at which f changes along the curve at t_0 . It is the rule $f \mapsto \dot{c}(t_0)(f)$, not any one of its numerical values, that represents the tangent vector.

Ordinary differentiation is linear and obeys the product rule. Writing $v = \dot{c}(t_0)$, these properties become

$$v(af + bh) = a v(f) + b v(h), \quad v(fh) = v(f)h(p) + f(p)v(h),$$

where $a, b \in \mathbb{R}$. The values $f(p)$ and $h(p)$ appear because the derivative is evaluated at the parameter value for which $c(t_0) = p$. These properties will also define tangent vectors without first choosing a curve.

3.2 Local functions and tangent vectors

Only the behavior of f near p matters in (3.1). To avoid requiring a function to be defined on all of M , consider pairs (U, f) , where U is an open neighborhood of p and $f \in C^\infty(U)$. Two pairs (U, f) and (W, h) are identified when $f = h$ on some open neighborhood of p contained in $U \cap W$. The resulting equivalence class is the *germ of f at p* , denoted by $[f]_p$. Equality of germs means agreement on a neighborhood, not just equality of the two values at p .

The set of smooth germs is denoted by $C_p^\infty(M)$. Addition and multiplication are performed after restricting representatives to a common neighborhood. These operations do not depend on the representatives, so $C_p^\infty(M)$ is a real algebra. Evaluation $[f]_p \mapsto f(p)$ is also well defined. We usually write $v(f)$ instead of $v([f]_p)$.

Definition 3.1: Tangent vector

A *tangent vector at p* is an $\mathbb{R}$ -linear map

$$v : C_p^\infty(M) \longrightarrow \mathbb{R}$$

satisfying the Leibniz rule

$$v(fh) = v(f)h(p) + f(p)v(h).$$

Such a map is called a *derivation at p* . The space of all these derivations is the *tangent space* $T_p M$. Its vector-space operations are defined by

$$(v + w)(f) = v(f) + w(f), \quad (av)(f) = a v(f).$$

Equation (3.1) therefore defines an element of $T_p M$. The coordinate description below will show that every element of $T_p M$ can be obtained from a curve. Every derivation annihilates constants. Indeed, $v(1) = v(1 \cdot 1) = 2v(1)$, so $v(1) = 0$, and linearity then gives $v(a) = 0$ for each constant $a \in \mathbb{R}$.

3.3 The coordinate basis

SOURCE SLIDES 9–10

Let (U, φ) be a smooth chart about p , with coordinate functions $x^1, \dots, x^n$. Write $u^1, \dots, u^n$ for the ordinary Cartesian coordinates on the image $\varphi(U) \subset \mathbb{R}^n$. Thus $x^i = u^i \circ \varphi$. A function f on U is represented in these coordinates by $\tilde{f} = f \circ \varphi^{-1}$.

Differentiate this representative with respect to its i -th coordinate and evaluate at $\varphi(p)$. This defines an operator on functions,

$$\partial_i|_p(f) := \left. \frac{\partial(f \circ \varphi^{-1})}{\partial u^i} \right|_{\varphi(p)}, \quad \partial_i = \frac{\partial}{\partial x^i}.$$

The notation $\partial_i f(p)$ abbreviates the same value. Since a partial derivative is linear and satisfies the product rule, $\partial_i|_p$ is a tangent vector. Applying it to a coordinate function gives

$$\partial_i|_p(x^j) = \delta_i^j.$$

Theorem 3.2: Coordinate basis of the tangent space

The vectors $\partial_1|_p, \dots, \partial_n|_p$ form a basis of $T_p M$. Every tangent vector has a unique expansion

$$v = v^i \partial_i|_p, \quad v^i = v(x^i), \quad v(f) = v^i \partial_i f(p). \quad (3.2)$$

In particular, $\dim T_p M = \dim M = n$.

Proof. If $b^i \partial_i|_p = 0$, applying both sides to x^j gives $b^j = 0$. Hence the coordinate vectors are linearly independent.

To prove that they span $T_p M$, we must show that a derivation is completely determined by its values on the coordinate functions. Let v be any derivation at p and put $a = \varphi(p)$. Restrict to a convex coordinate neighborhood of a . The fundamental theorem of calculus along the segment from a to u expresses the change in f in terms of the coordinate increments,

$$\tilde{f}(u) - \tilde{f}(a) = (u^i - a^i) h_i(u), \quad h_i(u) = \int_0^1 \frac{\partial \tilde{f}}{\partial u^i}(a + t(u - a)) dt.$$

The functions h_i are smooth, and $h_i(a) = \partial_i f(p)$. On the corresponding neighborhood of p , this identity reads

$$f - f(p) = (x^i - a^i)(h_i \circ \varphi).$$

Apply v and use the Leibniz rule. Constants are annihilated, and $x^i(p) - a^i = 0$, so

$$\begin{aligned} v(f) &= v(x^i) h_i(a) + (x^i(p) - a^i) v(h_i \circ \varphi) \\ &= v(x^i) \partial_i f(p). \end{aligned}$$

This is the action of $v(x^i) \partial_i|_p$ on every germ. Hence it is v . Uniqueness follows from linear independence. $\square$

The integral identity is exact for smooth functions. It does not assume that f is analytic or equal to a Taylor series. The theorem shows that a derivation is determined by the n numbers $v(x^i)$, although it acts on an infinite-dimensional algebra of functions. Only first partial derivatives of f enter its value.

3.4 Recovering the velocity of a curve

Write $x^i(t) = x^i(c(t))$. The ordinary chain rule gives

$$\left. \frac{d}{dt} \right|_{t=t_0} f(c(t)) = \dot{x}^i(t_0) \partial_i f(p), \quad p = c(t_0).$$

Comparing this with (3.2) yields

$$\dot{c}(t_0) = \dot{x}^i(t_0) \partial_i|_{c(t_0)}. \quad (3.3)$$

Thus the derivatives $\dot{x}^i(t_0)$ are the components of the tangent vector in the coordinate basis. They are not the basis vectors themselves.

Conversely, let $v = v^i \partial_i|_p$. For sufficiently small t , the curve

$$c(t) = \varphi^{-1}(\varphi(p) + t(v^1, \dots, v^n))$$

is defined, passes through p at $t = 0$, and has $\dot{c}(0) = v$. The straight line used in this formula lies in the coordinate image. Its inverse image need not look straight in an embedding of the manifold. This construction proves that every tangent vector is a curve velocity.

Taking the component tuple to be the i -th standard vector of $\mathbb{R}^n$ gives a curve whose velocity is $\partial_i|_p$. Along that curve, only x^i changes, at unit parameter rate. Two curves through p determine the same tangent vector precisely when their coordinate velocities agree in one chart, and hence in every chart.

For the helix $c(t) = (r \cos t, r \sin t, bt)$ in $\mathbb{R}^3$, with fixed $r > 0$ and $b \in \mathbb{R}$, the usual velocity components are $(-r \sin t, r \cos t, b)$. As a tangent derivation, the same velocity is

$$\dot{c}(t) = -r \sin t \partial_x|_{c(t)} + r \cos t \partial_y|_{c(t)} + b \partial_z|_{c(t)}.$$

Its action on f is the corresponding linear combination of partial derivatives of f , evaluated at $c(t)$.

Under a regular change of parameter $\tilde{c}(s) = c(t(s))$,

$$\tilde{c}'(s_0) = t'(s_0) \dot{c}(t(s_0)).$$

Regularity means that the parameter change is a local diffeomorphism, so $t'(s_0) \neq 0$. For a nonzero velocity, a positive factor preserves its oriented tangent direction and a negative factor reverses it. Changing the parameter can therefore change the tangent vector even when the curve traces the same set of points.

The velocity records the first-order behavior of a curve at the point. The following notation lets us compare that behavior for two smooth curves. For a vector-valued function $h(t)$, the notation $h(t) = O(t^2)$ as $t \rightarrow 0$ means that $\|h(t)\| \leq C|t|^2$ for some constant C and all sufficiently small t .

Student work – first-order equivalence of curves

Let $c_1(0) = c_2(0) = p$ for two smooth curves. Prove that $\dot{c}_1(0) = \dot{c}_2(0)$ if and only if

$$\varphi(c_1(t)) - \varphi(c_2(t)) = O(t^2) \quad (t \rightarrow 0)$$

in one chart about p . Show that the condition then holds in every chart. Give two parametrizations of the same curve with different nonzero velocities at p .

4 Differentials and coordinate transformations

SOURCE SLIDES 9–10

4.1 The differential of a map

Let $F : M \rightarrow N$ be smooth. A curve c through p is sent to the curve $F \circ c$ through $F(p)$. The *differential* of F relates their tangent vectors. Its input lies in $T_p M$, and its output lies in $T_{F(p)} N$.

To define the output as a derivation, let f be a smooth function near $F(p)$. The composition $f \circ F$ is a smooth function near p , so a vector $v \in T_p M$ can act on it.

Definition 4.1: Differential or pushforward

For a smooth map $F : M \rightarrow N$, its *differential at p* is the linear map

$$dF_p : T_p M \longrightarrow T_{F(p)} N, \quad (dF_p v)(f) = v(f \circ F).$$

It is also called the *pushforward at p* . The function f represents a germ at $F(p)$.

If two representatives of f agree near $F(p)$, their compositions with F agree near p . Thus the definition is independent of the representative. The Leibniz rule gives

$$(dF_p v)(fh) = (dF_p v)(f) h(F(p)) + f(F(p)) (dF_p v)(h),$$

so the output is a tangent vector at $F(p)$. Linearity in v follows directly from the definition.

When $v = \dot{c}(0)$, its action is

$$(dF_p v)(f) = \left. \frac{d}{dt} \right|_{t=0} f(F(c(t))).$$

This is the action of the velocity of $F \circ c$. Hence

$$dF_p(\dot{c}(0)) = (F \circ c)'(0).$$

Choose coordinates x^i near p and y^a near $F(p)$. The functions $F^a = y^a \circ F$ give the target coordinates of the image point as functions on the domain. Applying the differential to the coordinate basis gives

$$dF_p(\partial_i|_p) = \frac{\partial F^a}{\partial x^i}(p) \frac{\partial}{\partial y^a} \Big|_{F(p)}. \quad (4.1)$$

Indeed, the a -th component of the left-hand side is its value on y^a , namely $\partial_i(y^a \circ F)(p)$. The Jacobian is therefore the matrix of dF_p in these two coordinate bases. It need not be square when the manifolds have different dimensions.

For smooth maps $F : M \rightarrow N$ and $G : N \rightarrow P$, the chain rule is

$$d(G \circ F)_p = dG_{F(p)} \circ dF_p.$$

Both sides send v to the derivation $h \mapsto v(h \circ G \circ F)$. If F is a diffeomorphism, applying this identity to $F^{-1} \circ F$ shows that dF_p is invertible, with inverse $d(F^{-1})_{F(p)}$.

4.2 Changing coordinates at a fixed point

Now consider two charts on the *same* manifold, with coordinates related by $y^a = y^a(x)$. The point p and the vector $v \in T_p M$ remain fixed. Only their coordinate descriptions change. The chain rule for partial derivatives gives

$$\frac{\partial}{\partial y^a} \Big|_p = \frac{\partial x^i}{\partial y^a}(p) \partial_i|_p, \quad v'^a = \frac{\partial y^a}{\partial x^i}(p) v^i. \quad (4.2)$$

The second formula follows by writing v in both bases, or by applying v to y^a . The basis and its component column transform inversely, as in SR1. A Lorentz coordinate change has a constant Jacobian and gives the familiar law $v'^\mu = \Lambda^\mu_\nu v^\nu$.

A coordinate change here is not a map moving the event p to a new event. Its Jacobian converts between bases of one tangent space. By contrast, the differential of a general map F sends a vector at p to a vector at $F(p)$.

The first derivative of a curve transforms by the vector law. A second differentiation also differentiates the Jacobian. Along the curve,

$$\ddot{y}^a = \frac{\partial y^a}{\partial x^i} \ddot{x}^i + \frac{\partial^2 y^a}{\partial x^i \partial x^j} \dot{x}^i \dot{x}^j. \quad (4.3)$$

All derivatives of the coordinate transition are evaluated at the current point on the curve. The second term prevents $\ddot{x}^i$ from being the components of a vector under general coordinate changes. A connection will provide the correction needed to define covariant acceleration.

Student work – second derivatives under a coordinate change

Derive (4.3) and apply it to $x^1(t) = t$, $x^2(t) = 0$, with $y^1 = x^1$, $y^2 = x^2 + (x^1)^2$. Then derive the transformation law of $\partial_i \partial_j f$ for a scalar field and determine why it is not generally a covariant two-tensor.

5 Cotangent spaces, bundles, and fields

SOURCE SLIDE 10

5.1 Covectors and the differential of a function

The tangent space $T_p M$ is a vector space, so its dual is defined just as in SR1. The *cotangent space* is

$$T_p^* M = (T_p M)^* = \text{Hom}_{\mathbb{R}}(T_p M, \mathbb{R}).$$

Its elements are covectors at p . A covector takes a tangent vector as its input and returns a real number. This differs from a tangent vector viewed as a derivation, whose input is a smooth function germ.

Definition 5.1: Differential of a scalar field

For a smooth function f defined near p , its *differential at p* is the covector

$$(df)_p(v) = v(f), \quad v \in T_p M.$$

For coordinate functions $x^1, \dots, x^n$, their differentials satisfy

$$dx^i|_p(\partial_j|_p) = \delta^i_j, \quad (df)_p = \partial_i f(p) dx^i|_p. \quad (5.1)$$

Thus $dx^1|_p, \dots, dx^n|_p$ form the dual coordinate basis.

The first identity follows by evaluating $\partial_j|_p$ on x^i . For the second, both sides evaluated on $v = v^j \partial_j|_p$ give $v^j \partial_j f(p)$, so they are the same covector. In particular, $dx^i|_p$ extracts the component v^i of a tangent vector. For a curve velocity,

$$(df)_p(\dot{c}(t_0)) = \left. \frac{d}{dt} \right|_{t=t_0} f(c(t)), \quad c(t_0) = p.$$

The differential records all directional rates of change of f at the point. It is not tied to one particular curve.

This agrees with the differential of the map $f : M \rightarrow \mathbb{R}$ from the previous section. That differential takes values in $T_{f(p)} \mathbb{R}$, which is canonically identified with $\mathbb{R}$ by its usual coordinate. Under this identification it is precisely the covector $(df)_p$.

SOURCE SLIDE 11

For $v = v^i \partial_i|_p$ and $\omega = \omega_i dx^i|_p$, evaluation gives $\omega(v) = \omega_i v^i$. Under $y^a = y^a(x)$,

$$dy^a = \frac{\partial y^a}{\partial x^i} dx^i, \quad \omega'_a = \frac{\partial x^i}{\partial y^a} \omega_i. \quad (5.2)$$

The first relation follows from (5.1). Expressing ω in both dual bases gives the second. Together with the vector transformation law, it gives $\omega'_a v'^a = \omega_i v^i$.

Evaluation also gives the canonical identification

$$T_p M \longrightarrow T_p^{**} M, \quad v \longmapsto [\omega \longmapsto \omega(v)].$$

This identifies a vector with a linear functional on *covectors*. It does not identify $T_p M$ with $T_p^* M$. A metric will later provide that additional identification. In particular, df is a covector field, whereas its gradient will be a metric-dependent vector field.

5.2 Assembling the tangent spaces

To describe vectors at all points of M , we must retain both the base point and its tangent vector. The tangent bundle collects these data in one space. Vectors based at different points remain distinct elements, even when their coordinate component lists happen to agree.

Definition 5.2: Tangent and cotangent bundles

The *tangent bundle* and *cotangent bundle* are the disjoint unions

$$TM = \bigsqcup_{p \in M} T_p M, \quad T^* M = \bigsqcup_{p \in M} T_p^* M.$$

An element $v_p \in TM$ records a point p and a vector in $T_p M$. The *bundle projection* sends this element to its base point, $\pi(v_p) = p$. The set $\pi^{-1}(p) = T_p M$ is the *fiber* over p . The cotangent projection and its fibers are defined in the same way.

A chart (U, φ) , with coordinates $x^1, \dots, x^n$, supplies coordinates on $\pi^{-1}(U) \subset TM$. Expand $v_p = v^i \partial_i|_p$ and assign the tuple

$$v_p \longmapsto (x^1(p), \dots, x^n(p), v^1, \dots, v^n).$$

The first n numbers specify the point. The remaining n numbers specify the vector in the coordinate basis at that point. For $\omega_p = \omega_i dx^i|_p$, the analogous cotangent coordinates are $(x^1(p), \dots, x^n(p), \omega_1, \dots, \omega_n)$.

On an overlap, the induced coordinate changes are

$$\begin{aligned} (x^i, v^i) &\longmapsto \left(y^a(x), \frac{\partial y^a}{\partial x^i}(x) v^i \right), \\ (x^i, \omega_i) &\longmapsto \left(y^a(x), \frac{\partial x^i}{\partial y^a}(y(x)) \omega_i \right). \end{aligned}$$

Both are smooth, and both are linear in the vector or covector components when the base point is fixed. They equip TM and T^*M with smooth manifold structures. Their dimensions are

$$\dim T_p M = \dim T_p^* M = n, \quad \dim TM = \dim T^* M = 2n.$$

The dimension $2n$ counts both position and fiber coordinates. An individual fiber still has dimension n .

Both bundles are locally products, and their changes of fiber coordinates are linear. The following definition isolates this common structure.

Definition 5.3: Smooth vector bundle

A smooth *vector bundle of rank r* over M consists of a smooth manifold E , a smooth surjection $\pi : E \rightarrow M$, and an r -dimensional vector-space structure on each fiber $E_p = \pi^{-1}(p)$. Every $p \in M$ has an open neighborhood U and a diffeomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^r$$

that leaves the base point unchanged and restricts to a linear isomorphism $E_q \rightarrow \{q\} \times \mathbb{R}^r$ for every $q \in U$. The map Φ is a *local trivialization*.

A local trivialization represents each fiber over U by a copy of $\mathbb{R}^r$, with the representation varying smoothly with the point. The constructions above make TM and T^*M rank- n vector bundles. The existence of these local product descriptions does not assert that either bundle is a product over all of M .

5.3 Sections and tensor fields

A field chooses one element from each fiber. It is described by a map into the total space whose projection returns the point where the field is evaluated.

Definition 5.4: Section, vector field, and one-form

A *section* of $\pi : E \rightarrow M$ is a map $s : M \rightarrow E$ satisfying $\pi \circ s = \text{id}_M$. The space of smooth sections is denoted by $\Gamma(E)$.

A *smooth vector field* is a smooth section of TM , and a *smooth one-form* is a smooth section of T^*M . Their spaces are

$$\mathfrak{X}(M) = \Gamma(TM), \quad \Omega^1(M) = \Gamma(T^*M).$$

Locally, a vector field and a one-form have the expressions $X = X^i \partial_i$ and $\alpha = \alpha_i dx^i$. Smoothness means that their coefficient functions are smooth. A tangent vector X_p differentiates a function to a number. A vector field performs this operation at every point, producing another smooth function,

$$X(f)(p) = X_p(f), \quad X(f) = X^i \partial_i f.$$

It obeys $X(fh) = X(f)h + fX(h)$.

Fields can be added and multiplied by real constants. They can also be multiplied by smooth functions, with $(fX)_p = f(p)X_p$. Thus $\mathfrak{X}(M)$ is a real vector space and a *module over $C^\infty(M)$* . Here “module” means that the multipliers may be smooth functions rather than only real numbers. Linearity with respect to such multipliers is stronger than real linearity. This distinction will be essential when defining derivatives of fields.

A *local frame* consists of smooth vector fields $e_1, \dots, e_n$ that form a basis at every point of an open set. Their pointwise dual one-forms form the *dual coframe*. Coordinate fields give one example of a local frame. An arbitrary smooth frame need not arise from coordinates.

A diffeomorphism $F : M \rightarrow N$ pushes a vector field forward by

$$(F_*X)_{F(p)} = dF_p(X_p).$$

Each target point has exactly one preimage, so this defines a smooth field on N . For a general smooth map, different preimages may produce different vectors at the same target point. The pointwise differential therefore does not automatically define a vector field on the target.

More generally, a tensor field of type (r, s) is a smooth section of

$$T_s^r M = (TM)^{\otimes r} \otimes (T^*M)^{\otimes s}.$$

The fiber over p is $(T_p M)^{\otimes r} \otimes (T_p^* M)^{\otimes s}$. All its factors refer to the same point. Its components obey the tensor transformation law from SR1, with the coordinate Jacobian replacing the constant Lorentz matrix.

For example, a smooth family of linear maps $A_p : T_p M \rightarrow T_p M$ is a tensor field of type $(1, 1)$, written

$$A = A^i_j \partial_i \otimes dx^j, \quad A(X) = A^i_j X^j \partial_i.$$

Under $y^a = y^a(x)$, its components at the same point satisfy

$$A'^a_b(p) = \frac{\partial y^a}{\partial x^i}(p) \frac{\partial x^j}{\partial y^b}(p) A^i_j(p).$$

Here A^i_j and A'^a_b denote the component functions in the two charts. The identity map has components δ^i_j in every chart.

A useful test for tensoriality is linearity over $C^\infty(M)$ in each field argument. Expand the fields in a local basis. Such linearity allows their component functions to be taken outside the operation, without differentiating them. For example, if a smooth operation $B(X, Y)$ has this property in both arguments, then locally

$$B(X, Y)_p = X^i(p) Y^j(p) B(\partial_i, \partial_j)_p.$$

Its value depends only on X_p and Y_p , not on derivatives of the fields. A scalar-valued operation of this kind is a covariant two-tensor. A vector-valued one is a tensor of type $(1, 2)$.

6 Tangent geometry of the two-sphere

SOURCE SLIDE 12

The sphere lets us compare the intrinsic definitions with familiar vectors in three-dimensional space. Its tangent vectors can be represented by ambient arrows, provided their base points are retained.

6.1 Intrinsic vectors and the ambient tangent plane

Let $\iota : S^2 \hookrightarrow \mathbb{R}^3$ be the inclusion. A curve on the sphere is also a curve in $\mathbb{R}^3$, and $d\iota_p$ sends its intrinsic velocity to its ambient velocity. Using Cartesian coordinates to identify $T_{\iota(p)}\mathbb{R}^3$ with $\mathbb{R}^3$, we obtain the usual tangent plane as a vector subspace.

Proposition 6.1: Tangent space of the unit sphere

For every $p \in S^2$,

$$d\iota_p(T_p S^2) = p^\perp = \{v \in \mathbb{R}^3 : p \cdot v = 0\}.$$

With this identification, the tangent bundle can be written as

$$TS^2 = \{(p, v) \in \mathbb{R}^3 \times \mathbb{R}^3 : p \cdot p = 1, p \cdot v = 0\}.$$

Proof. If $c(0) = p$ is a curve on the sphere, differentiating $c(t) \cdot c(t) = 1$ gives $p \cdot \dot{c}(0) = 0$. Conversely, for $v \in p^\perp$, the curve

$$c(t) = \frac{p + tv}{\sqrt{1 + t^2 \|v\|^2}}$$

lies on the sphere and has $c(0) = p$, $\dot{c}(0) = v$. This also covers $v = 0$. The image of $d\iota_p$ is therefore the two-dimensional space $p^\perp$. Since $\dim T_p S^2 = 2$, the differential is an isomorphism onto this image. $\square$

The affine tangent plane drawn through the point is $p + p^\perp$. The vector space representing tangent vectors is $p^\perp$. Keeping the base point in the pair (p, v) distinguishes the fibers even when their ambient vector representations happen to coincide.

6.2 Spherical coordinates and coordinate vectors

On the sphere away from the poles and a chosen longitude cut, use

$$\sigma(\vartheta, \varphi) = (\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta),$$

where $0 < \vartheta < \pi$ and $\varphi_0 < \varphi < \varphi_0 + 2\pi$. The inverse of this parametrization is the chart. The ambient representatives of its coordinate vectors are

$$e_\vartheta = \frac{\partial \sigma}{\partial \vartheta} = (\cos \vartheta \cos \varphi, \cos \vartheta \sin \varphi, -\sin \vartheta), \quad (6.1)$$

$$e_\varphi = \frac{\partial \sigma}{\partial \varphi} = \sin \vartheta (-\sin \varphi, \cos \varphi, 0). \quad (6.2)$$

These represent ∂_ϑ and ∂_φ , respectively. The coordinate vectors are tangent and linearly independent on the stated chart. Hence

$$V = V^\vartheta \partial_\vartheta + V^\varphi \partial_\varphi$$

has ambient representative $V^\vartheta e_\vartheta + V^\varphi e_\varphi$. The tuple $(\vartheta, \varphi, V^\vartheta, V^\varphi)$ gives local coordinates on TS^2 .

At either pole, the formal expression for e_φ vanishes. The tangent space has not lost a dimension. Longitude is not a coordinate at a pole, so this particular chart and its frame do not apply there.

Student work – a global vector field and a local coordinate frame

For $K(x, y, z) = (-y, x, 0)$, prove tangency to S^2 , find its angular coordinate components, and locate its zeros. Explain why K is smooth at the poles although the angular coordinate frame is not defined there.

6.3 Comparing vectors at different points

SOURCE SLIDE 13

Changing the basis at p keeps the vector in the same space $T_p S^2$. Comparing a vector at p with one at another point q is a different operation. The second vector belongs to $T_q S^2$, so $V_q - V_p$ is not defined by the manifold structure alone.

The embedding lets us subtract the ambient representatives in $\mathbb{R}^3$. But differentiating a tangent field in $\mathbb{R}^3$ need not give a tangent vector. Its derivative may contain a component normal to the sphere. To obtain a derivative within the tangent spaces, a rule for comparing the fibers along a curve must be specified. This is the role of a connection.

7 Connections and parallel transport

SOURCE SLIDE 14

7.1 Covariant differentiation

For a scalar field, nearby values can be subtracted because they all belong to $\mathbb{R}$. For a vector field Y , the values Y_p and Y_q belong to different tangent spaces. A derivative must specify how these values are compared. An affine connection supplies such a rule.

Differentiating the component functions alone does not solve the problem. If $Y'^a = (\partial y^a / \partial x^j) Y^j$, then

$$\partial'_b Y'^a = \frac{\partial x^i}{\partial y^b} \frac{\partial y^a}{\partial x^j} \partial_i Y^j + \frac{\partial x^i}{\partial y^b} \frac{\partial^2 y^a}{\partial x^i \partial x^j} Y^j.$$

The second term comes from differentiating the change-of-basis matrix. It is absent from a tensor transformation law. A connection adds terms that compensate for this dependence on the chosen frame. Choosing the connection is additional geometric data, not just a change of notation.

Definition 7.1: Affine connection

An *affine connection* on TM is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \longrightarrow \mathfrak{X}(M), \quad (X, Y) \longmapsto \nabla_X Y,$$

satisfying

$$\begin{aligned} \nabla_{fX+hY} Z &= f\nabla_X Z + h\nabla_Y Z, \\ \nabla_X(aY + bZ) &= a\nabla_X Y + b\nabla_X Z, \\ \nabla_X(fY) &= X(f)Y + f\nabla_X Y, \end{aligned}$$

for smooth fields X, Y, Z , functions $f, h \in C^\infty(M)$, and constants $a, b \in \mathbb{R}$. The field $\nabla_X Y$ is the *covariant derivative of Y along X* .

The first identity is linearity over $C^\infty(M)$ in X . At a point, only the vector X_p is needed to specify the differentiation direction. The second and third identities say that differentiation in Y is real-linear and obeys a product rule. It may therefore depend on the first derivatives of Y , not just its value at the point.

To work in a chart, we need the derivative to depend only on the fields near the point. This locality follows from the axioms. Suppose Y vanishes near p . Choose a smooth cutoff χ equal to one near p , with support inside that neighborhood. It can be constructed in a coordinate ball and extended by zero. Since $\chi Y = 0$, the product rule at p gives $(\nabla_X Y)_p = 0$. First-slot linearity gives the same conclusion when X vanishes near p . Thus local fields may be extended using cutoffs, and the derivative near p is independent of the extension. We can therefore apply ∇ to local coordinate fields.

Define the *connection coefficients* by

$$\nabla_{\partial_i} \partial_j = \Gamma^k_{ij} \partial_k. \quad (7.1)$$

The first lower index is the differentiation direction. The second identifies the basis vector being differentiated. For $X = X^i \partial_i$ and $Y = Y^j \partial_j$, the axioms give

$$\begin{aligned} \nabla_X Y &= X^i \nabla_{\partial_i} (Y^j \partial_j) \\ &= X^i (\partial_i Y^j) \partial_j + X^i Y^j \Gamma^k_{ij} \partial_k. \end{aligned}$$

Renaming the summed indices yields

$$\nabla_X Y = X^i (\partial_i Y^k + \Gamma^k_{ij} Y^j) \partial_k. \quad (7.2)$$

The first term differentiates the components. The second accounts for the covariant derivatives of the basis vectors. Their sum is a vector, although the two terms do not separately transform as vectors.

For fixed Y , the map $X \mapsto \nabla_X Y$ is linear over $C^\infty(M)$. It therefore defines a tensor field of type $(1, 1)$,

$$\nabla Y = (\partial_i Y^k + \Gamma^k_{ij} Y^j) \partial_k \otimes dx^i.$$

The notation $\nabla_i Y^k$ denotes the component in parentheses. It is not the ordinary derivative of the single scalar function Y^k .

Proposition 7.2: Transformation of connection coefficients

For the coordinate change $y^a = y^a(x)$, the coefficients of the same connection satisfy

$$\tilde{\Gamma}^a_{bc} = \frac{\partial y^a}{\partial x^k} \frac{\partial x^i}{\partial y^b} \frac{\partial x^j}{\partial y^c} \Gamma^k_{ij} + \frac{\partial y^a}{\partial x^k} \frac{\partial^2 x^k}{\partial y^b \partial y^c}. \quad (7.3)$$

The second term shows that connection coefficients are not the components of a tensor.

To obtain this law, substitute $\partial'_c = (\partial x^j / \partial y^c) \partial_j$ into $\nabla_{\partial'_c} \partial'_c$. Differentiating the coefficient $\partial x^j / \partial y^c$ produces the second-derivative term. Expressing the result in the primed basis gives (7.3). This extra term cancels the unwanted term from the transformation of $\partial_i Y^k$.

7.2 Differentiation along a curve

Let $\gamma : [a, b] \rightarrow M$ be a smooth curve. A *vector field along* γ assigns a vector $V(t) \in T_{\gamma(t)}M$, with smooth components in local coordinates. Write $V(t) = V^k(t) \partial_k|_{\gamma(t)}$. This need not come from one field on M . For example, a curve can pass through the same point at two parameter values and carry different vectors on the two visits.

The same product rule used for $\nabla_X Y$ now differentiates $V^k(t)$, while the basis varies along γ . The direction components X^i are replaced by the curve velocity $\dot{\gamma}^i$.

Definition 7.3: Covariant derivative along a curve

The connection defines

$$\frac{DV}{dt} = \left(\frac{dV^k}{dt} + \Gamma^k_{ij}(\gamma(t)) \dot{\gamma}^i V^j \right) \partial_k|_{\gamma(t)}. \quad (7.4)$$

The field V is *parallel along* γ when $DV/dt = 0$.

The transformation law of the connection coefficients makes this expression coordinate independent. If $V(t) = Y_{\gamma(t)}$ for a local vector field Y , it agrees with $\nabla_{\dot{\gamma}} Y$. Taking $V = \dot{\gamma}$ gives the covariant acceleration,

$$\frac{D\dot{\gamma}}{dt} = (\ddot{\gamma}^k + \Gamma^k_{ij} \dot{\gamma}^i \dot{\gamma}^j) \partial_k|_{\gamma(t)}.$$

The connection term corrects the nonvectorial transformation of ordinary coordinate acceleration found in Section 4.

Theorem 7.4: Parallel transport along a curve

For each $v \in T_{\gamma(a)}M$, there is a unique parallel field along γ satisfying $V(a) = v$. Its endpoint value defines a linear isomorphism

$$P_\gamma : T_{\gamma(a)}M \longrightarrow T_{\gamma(b)}M, \quad P_\gamma(v) = V(b),$$

called *parallel transport along* γ .

Proof. In a chart, the parallel condition is the linear system of ordinary differential equations

$$\dot{V}^k = -\Gamma^k_{ij}(\gamma(t)) \dot{\gamma}^i V^j.$$

Existence and uniqueness hold on each compact subinterval contained in one chart. A finite subdivision of $[a, b]$ allows these solutions to be continued along the whole curve and joined uniquely

on overlaps. Linearity of the equations makes the endpoint map linear. For the reversed curve $\bar{\gamma}(t) = \gamma(a + b - t)$, the evolution is reversed, giving $P_{\bar{\gamma}} = P_{\gamma}^{-1}$. $\square$

Parallel transport now provides a comparison of vectors in different tangent spaces. To compare $v \in T_{\gamma(a)}M$ with $w \in T_{\gamma(b)}M$, first transport v . The difference $w - P_{\gamma}v$ is defined because both terms lie in the endpoint tangent space. The result can depend on the chosen path, not only on its endpoints.

The same comparison explains covariant differentiation. Fix t_0 and let $P_{t_0+h \rightarrow t_0}$ denote transport back along γ . For a field V along the curve,

$$\left. \frac{DV}{dt} \right|_{t_0} = \lim_{h \rightarrow 0} \frac{P_{t_0+h \rightarrow t_0} V(t_0 + h) - V(t_0)}{h}.$$

All vectors in this difference quotient belong to $T_{\gamma(t_0)}M$. Expanding the transport equation to first order in h recovers (7.4).

Example 7.5: Tangential differentiation on the sphere

For $\gamma(t) \in S^2$, represent a tangent field by an ambient vector satisfying $\gamma \cdot V = 0$. Using the Euclidean scalar product in $\mathbb{R}^3$, project its derivative onto the tangent plane,

$$\frac{DV}{dt} = \dot{V} - (\gamma \cdot \dot{V})\gamma = \dot{V} + (\dot{\gamma} \cdot V)\gamma.$$

The second equality follows by differentiating $\gamma \cdot V = 0$. This rule defines a connection on the sphere. Along the equator $\gamma(t) = (\cos t, \sin t, 0)$, the field $V = \dot{\gamma}$ has $\dot{V} = -\gamma$ and $DV/dt = 0$. Its ambient derivative is nonzero, but entirely normal to the sphere.

Example 7.6: The Euclidean connection in polar coordinates

Choose the connection on the plane for which the Cartesian basis fields are parallel. On a polar chart,

$$\partial_r = \cos \varphi \partial_x + \sin \varphi \partial_y, \quad \partial_\varphi = -r \sin \varphi \partial_x + r \cos \varphi \partial_y.$$

The product rule gives

$$\nabla_{\partial_r} \partial_r = 0, \quad \nabla_{\partial_r} \partial_\varphi = \nabla_{\partial_\varphi} \partial_r = \frac{1}{r} \partial_\varphi, \quad \nabla_{\partial_\varphi} \partial_\varphi = -r \partial_r.$$

Thus the nonzero coefficients are

$$\Gamma^r_{\varphi\varphi} = -r, \quad \Gamma^\varphi_{r\varphi} = \Gamma^\varphi_{\varphi r} = \frac{1}{r}.$$

The Cartesian coefficients vanish and the polar coefficients do not, but both describe the same connection.

Student work – parallelism and varying components

Verify the polar-coordinate derivatives above. Express ∂_x in the polar frame and show that $\nabla_{\partial_r}\partial_x = \nabla_{\partial_\varphi}\partial_x = 0$. Explain why a parallel field need not have constant coordinate components.

No metric was required in the definition of an affine connection. A metric selects a particular connection once metric compatibility and zero torsion are imposed. Appendix A develops these conditions after the metric and Lie bracket have been introduced.

8 Differential forms and exterior algebra

SOURCE SLIDE 15

A one-form acts linearly on a tangent vector. A differential form of higher degree acts multilinearly on several tangent vectors and is *alternating*, meaning that interchanging two arguments reverses its sign. All the arguments are vectors at the same point. This antisymmetry will allow a derivative in which the extra terms arising from coordinate changes cancel.

Definition 8.1: Differential form

For $k \geq 1$, the space $\Lambda^k T_p^* M$ consists of alternating multilinear maps

$$\alpha_p : (T_p M)^k \longrightarrow \mathbb{R}.$$

A *smooth differential k -form* assigns such a map to each $p \in M$, with smooth components in local coordinates. Equivalently, it is a smooth section of $\Lambda^k T^* M$. We write

$$\Omega^k(M) = \Gamma(\Lambda^k T^* M), \quad \Omega^0(M) = C^\infty(M).$$

Thus a zero-form is a smooth function.

8.1 The wedge product

For one-forms α and β , their *wedge product* is the two-form defined by

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v).$$

The subtraction makes the result alternating. In particular,

$$dx^i \wedge dx^j = -dx^j \wedge dx^i, \quad dx^i \wedge dx^i = 0.$$

More generally, the wedge product of k covectors at one point is

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) = \det(\alpha^i(v_j)).$$

Such products span $\Lambda^k T_p^* M$, but not every form is a single product of one-forms.

For $\alpha \in \Omega^k(M)$ and $\beta \in \Omega^\ell(M)$, evaluate the forms on separate groups of vectors, then antisymmetrize over all $k + \ell$ inputs. Write S_m for the permutations of $\{1, \dots, m\}$, and $\text{sgn}(\sigma)$ for the sign of a permutation σ . The resulting product is

$$(\alpha \wedge \beta)(v_1, \dots, v_{k+\ell}) = \frac{1}{k! \ell!} \sum_{\sigma \in S_{k+\ell}} \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ \times \beta(v_{\sigma(k+1)}, \dots, v_{\sigma(k+\ell)}).$$

Each form is already alternating in its own inputs. Permutations within these two groups therefore repeat each contribution $k! \ell!$ times, which explains the normalization. The formula defines a $(k + \ell)$ -form without first expressing the factors as sums of products of one-forms. Wedge multiplication is associative and satisfies

$$\alpha \wedge \beta = (-1)^{k\ell} \beta \wedge \alpha.$$

This is *graded commutativity*. The sign comes from moving k one-form factors past ℓ others, requiring $k\ell$ swaps. Degree-zero factors act by ordinary multiplication of a form by a function.

8.2 Components and normalization

The components of a k -form are defined by evaluation on coordinate basis vectors,

$$\alpha_{i_1 \dots i_k} = \alpha(\partial_{i_1}, \dots, \partial_{i_k}).$$

They are completely antisymmetric. The products $dx^{i_1} \wedge \dots \wedge dx^{i_k}$, with $i_1 < \dots < i_k$, form a basis of $\Lambda^k T_p^* M$ at each point. Consequently,

$$\alpha = \sum_{i_1 < \dots < i_k} \alpha_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \\ = \frac{1}{k!} \alpha_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}. \quad (8.1)$$

The second line sums over all index values. Permuting distinct indices changes the signs of both the component and the wedge product, so the same contribution occurs $k!$ times. This explains the factor $1/k!$. For example,

$$F = \frac{1}{2} F_{ij} dx^i \wedge dx^j, \quad F(\partial_i, \partial_j) = F_{ij}.$$

There are $\binom{n}{k}$ increasing index lists of length k . Hence

$$\dim \Lambda^k T_p^* M = \binom{n}{k}, \quad \dim \left(\bigoplus_{k=0}^n \Lambda^k T_p^* M \right) = 2^n.$$

These are dimensions of spaces of forms at *one point*. The space $\Omega^k(M)$ of smooth form fields is generally infinite-dimensional because its coefficients are functions. For $k > n$, alternating k -forms vanish: any k tangent vectors are linearly dependent.

No metric has been used. The determinant formula assigns a signed number once the covectors have been specified. It does not by itself supply the lengths or angles of the input vectors. Also, graded commutativity forces $\alpha \wedge \alpha = 0$ for odd-degree forms, but imposes no such condition for even degree.

Student work – alternating products in four dimensions

For $F = dx^1 \wedge dx^2 + dx^3 \wedge dx^4$ on $\mathbb{R}^4$, compute $F \wedge F$. Deduce that F cannot be written as $\alpha \wedge \beta$ for two one-forms.

9 Exterior differentiation and pullback

SOURCE SLIDE 16

9.1 The exterior derivative

For a scalar field, df is a one-form that records directional rates of change. The *exterior derivative* extends this operation to forms of every degree. It takes a k -form to a $(k+1)$ -form and requires neither a metric nor a connection.

Theorem 9.1: Exterior derivative

On a smooth manifold there is a unique family of real-linear operators

$$d : \Omega^k(M) \longrightarrow \Omega^{k+1}(M)$$

that agrees with the differential on functions and satisfies

$$d^2 = 0, \quad d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta, \quad \alpha \in \Omega^k(M).$$

In a chart, let $I = (i_1, \dots, i_k)$ range over increasing index lists and write $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k}$. For a coordinate expansion $\alpha = \sum_I \alpha_I dx^I$, its local formula is

$$d\alpha = \sum_I d\alpha_I \wedge dx^I = \sum_{I,j} \partial_j \alpha_I dx^j \wedge dx^I. \quad (9.1)$$

The local rule differentiates each coefficient and places its differential before the existing wedge factors. The coordinate one-forms contribute no extra terms because $d(dx^i) = d^2 x^i = 0$. For a one-form,

$$d(\alpha_i dx^i) = \partial_j \alpha_i dx^j \wedge dx^i.$$

Renaming the summed indices and using antisymmetry gives

$$d\alpha = \frac{1}{2}(\partial_i \alpha_j - \partial_j \alpha_i) dx^i \wedge dx^j. \quad (9.2)$$

Thus only the antisymmetric part of the coefficient derivatives remains. For $F = \frac{1}{2} F_{ij} dx^i \wedge dx^j$, the corresponding components are

$$(dF)_{ijk} = \partial_i F_{jk} + \partial_j F_{ki} + \partial_k F_{ij}.$$

Example 9.2: Exterior derivative of a one-form

On $\mathbb{R}^2$, let $\alpha = x \, dy - y \, dx$. Then

$$d\alpha = dx \wedge dy - dy \wedge dx = 2 \, dx \wedge dy.$$

For tangent vectors v, w , $(d\alpha)(v, w) = 2(v^x w^y - v^y w^x)$. The result is an alternating bilinear form.

To construct d , use (9.1) in each chart. The chain rule gives $dy^a = (\partial_i y^a) dx^i$ on an overlap. Applying the local rule once more gives

$$d(dy^a) = \partial_j \partial_i y^a \, dx^j \wedge dx^i = 0,$$

since the coefficient is symmetric in i, j and the wedge product is antisymmetric. The local operators therefore agree on functions and on coordinate differentials. The product rule makes them agree on every form. The same cancellation of mixed partial derivatives gives $d^2 = 0$.

For uniqueness, the product rule implies locality by a cutoff argument, as for connections. In a chart, the operator is fixed on functions and has $d(dx^i) = 0$. Applying its product rule to the coordinate expansion of any form then forces (9.1).

Student work – the identity $d^2 = 0$

Verify $d^2 f = 0$ directly from the coordinate formula for a smooth function. Extend the calculation to an arbitrary differential form.

A form is *closed* when $d\alpha = 0$. A form of positive degree is *exact* when $\alpha = d\beta$ for a globally defined form β . Every exact form is closed because $d^2 = 0$. The converse need not hold globally. An n -form on an n -manifold is always closed because there are no nonzero $(n+1)$ -forms. This does not imply that it is exact.

9.2 Pullback of forms

A smooth map $F : M \rightarrow N$ sends tangent vectors forward by its differential. A form on N can then evaluate these image vectors. This defines a form on M , called its *pullback*. The form moves in the opposite direction to the map, but no inverse of F is required.

Definition 9.3: Pullback

For $F : M \rightarrow N$ smooth and $\alpha \in \Omega^k(N)$, define $F^* \alpha \in \Omega^k(M)$ by

$$(F^* \alpha)_p(v_1, \dots, v_k) = \alpha_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

For a function, the definition is $F^* f = f \circ F$.

For a one-form, first send $v \in T_p M$ to $dF_p v$, then evaluate $\alpha_{F(p)}$ on that vector. In target coordinates y^a , it gives the practical formulas

$$F^*(dy^a) = d(y^a \circ F), \quad F^*(\alpha_a dy^a) = (\alpha_a \circ F) d(y^a \circ F).$$

Both the coefficients and the coordinate differentials must be pulled back. The active scalar-field transformation in SR1, $f \mapsto f \circ F^{-1}$, is therefore $(F^{-1})^* f$ when F is a diffeomorphism. It should not be confused with $F^* f$.

The same evaluation rule defines pullback for any covariant tensor. For a covariant two-tensor h ,

$$(F^* h)_p(v, w) = h_{F(p)}(dF_p v, dF_p w).$$

Even if h is nondegenerate, its pullback need not be. For example, a nonzero vector in the kernel of dF_p is paired to zero with every vector by $F^* h$.

Proposition 9.4: Pullback and exterior differentiation

For smooth maps and differential forms,

$$F^*(\alpha \wedge \beta) = F^* \alpha \wedge F^* \beta, \quad F^*(d\alpha) = d(F^* \alpha).$$

If $F : M \rightarrow N$ and $G : N \rightarrow P$, then $(G \circ F)^* = F^* \circ G^*$.

The wedge identity follows from the alternating evaluation rule, and the composition identity follows from the chain rule for differentials. For a function f , the chain rule gives $F^*(df) = d(f \circ F)$. Every form is locally a sum of terms $f dy^{i_1} \wedge \cdots \wedge dy^{i_k}$. Applying the product rules extends the identity to all forms. Thus pulling back and exterior differentiating can be performed in either order.

Example 9.5: A smooth one-form on the sphere

Let $\iota : S^2 \hookrightarrow \mathbb{R}^3$ be the inclusion and define

$$\beta = \iota^*(x dy - y dx).$$

Substituting the spherical parametrization gives

$$\beta = \sin^2 \vartheta d\varphi, \quad d\beta = 2 \sin \vartheta \cos \vartheta d\vartheta \wedge d\varphi.$$

The pullback definition gives a smooth one-form on the entire sphere. The angular expressions are valid only on their chart. In particular, $d\varphi$ itself is not defined at a pole, but this does not prevent β from being smooth there.

Student work – pullback to a surface

For $F(u, v) = (u, v, u^2 + v^2)$ and $\alpha = x \, dy + y \, dz + z \, dx$ on $\mathbb{R}^3$, compute $F^*\alpha$ and verify $d(F^*\alpha) = F^*(d\alpha)$ directly.

9.3 Local and global potentials

A potential for a one-form α is a smooth function f with $\alpha = df$. Integration tests whether such a potential can be defined globally. For an oriented smooth parametrized curve $\gamma : [a, b] \rightarrow M$,

$$\int_{\gamma} \alpha := \int_a^b \alpha_{\gamma(t)}(\dot{\gamma}(t)) \, dt.$$

This is the integral of the pullback $\gamma^*\alpha$ on the parameter interval. A change of parameter preserving orientation leaves it unchanged. For $\alpha = df$, the chain rule gives

$$\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a)).$$

In particular, an exact one-form integrates to zero around a closed curve.

On $\mathbb{R}^2 \setminus \{0\}$, consider

$$\alpha = \frac{-y \, dx + x \, dy}{x^2 + y^2}.$$

A coordinate calculation gives $d\alpha = 0$. In an angular chart, $\alpha = d\varphi$, so it has a local potential. But along the unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$,

$$\gamma^*\alpha = dt, \quad \int_{\gamma} \alpha = 2\pi.$$

It follows that no globally defined real-valued function f satisfies $\alpha = df$. The angle can be chosen smoothly on individual angular charts, but not as a single real-valued angle on the punctured plane. Thus local potentials do not always combine into a global one.

Exterior differentiation acts on alternating covariant tensors. The next construction instead uses the local flow of a vector field to differentiate vector fields and more general tensor fields, again without choosing a connection.

10 Vector fields, flows, and Lie derivatives

SOURCE SLIDE 17

10.1 The Lie bracket

A vector field X acts on smooth functions by $f \mapsto X(f)$. Given two fields X, Y , we can differentiate a function first with Y and then with X , obtaining $X(Y(f))$. Reversing the order need not give the same result. Their difference defines the Lie bracket.

Definition 10.1: Lie bracket

For smooth vector fields X, Y , the *Lie bracket* is the vector field determined by

$$[X, Y](f) = X(Y(f)) - Y(X(f)), \quad f \in C^\infty(M).$$

The definition must produce a first-order derivation, not a general second-order differential operator. To check this, expand in coordinates,

$$\begin{aligned} X(Y(f)) - Y(X(f)) &= (X^i \partial_i Y^k - Y^i \partial_i X^k) \partial_k f \\ &\quad + (X^i Y^j - Y^i X^j) \partial_i \partial_j f. \end{aligned}$$

The coefficient of the second derivatives is antisymmetric in i, j , while $\partial_i \partial_j f$ is symmetric. Their contraction vanishes, leaving

$$[X, Y] = (X^i \partial_i Y^k - Y^i \partial_i X^k) \partial_k. \quad (10.1)$$

The resulting coefficients are smooth. Since the original definition uses the actions of the fields on functions, the vector field does not depend on a coordinate choice.

Proposition 10.2: The Lie algebra of vector fields

The Lie bracket is real-bilinear and antisymmetric, and satisfies the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Thus $\mathfrak{X}(M)$, with this bracket, is a real Lie algebra.

A Lie algebra is a vector space with a bilinear bracket satisfying antisymmetry and the Jacobi identity. Here that vector space is $\mathfrak{X}(M)$, the space of vector fields, not the set of points of M . The bracket also satisfies

$$[X, fY] = X(f)Y + f[X, Y], \quad [fX, Y] = f[X, Y] - Y(f)X. \quad (10.2)$$

It is therefore not linear over $C^\infty(M)$ in either field argument. The value $[X, Y]_p$ depends on how the fields vary near p , not just on X_p and Y_p . Coordinate fields commute, $[\partial_i, \partial_j] = 0$, but general frame fields need not.

Student work – commutators and the Jacobi identity

Prove the Jacobi identity and (10.2). Explain why the Lie bracket is not a tensor of type $(1, 2)$.

10.2 Local flows

A vector field specifies a tangent vector at each point. An integral curve follows these vectors, with its velocity equal to the field at its current position. In coordinates, this means solving

$$\frac{dx^i(\gamma(t))}{dt} = X^i(\gamma(t)).$$

Solving this equation for different initial points produces a family of maps. The map at parameter value t sends each initial point to the point reached after following the field for that parameter interval.

Definition 10.3: Integral curve and local flow

An *integral curve* of X is a smooth curve $\gamma : I \rightarrow M$ satisfying $\dot{\gamma}(t) = X_{\gamma(t)}$.

The *local flow* of X is a smooth map $\Phi : \mathcal{D} \rightarrow M$, where $\mathcal{D}$ is an open neighborhood of $\{0\} \times M$ in $\mathbb{R} \times M$, such that

$$\Phi_0(p) = p, \quad \frac{d}{dt}\Phi_t(p) = X_{\Phi_t(p)}, \quad \Phi_t(p) = \Phi(t, p).$$

For each p , the allowed parameter values form an open interval containing zero. The curve $t \mapsto \Phi_t(p)$ is the integral curve starting at p .

Existence, uniqueness, and smooth dependence on initial conditions follow from the corresponding ordinary differential-equation results. Uniqueness gives

$$\Phi_{t+s}(p) = \Phi_t(\Phi_s(p)), \quad \Phi_{-t} = \Phi_t^{-1}$$

on the relevant common domains. Following the field for s and then for t is the same as following it for $t + s$. Each fixed-time map is a diffeomorphism between its domain and image.

A local flow need not exist for every real parameter value. For example, $X = x^2\partial_x$ on $\mathbb{R}$ gives $\Phi_t(a) = a/(1 - at)$, whose denominator vanishes at $t = 1/a$ when $a \neq 0$. A field is *complete* when every maximal integral curve is defined for all real times. The derivatives below use only parameter values near zero.

For the sphere field $K(x, y, z) = (-y, x, 0)$, the flow is the rotation

$$\Phi_s(x, y, z) = (x \cos s - y \sin s, x \sin s + y \cos s, z).$$

In angular coordinates, $K = \partial_\varphi$ and the same map is $(\vartheta, \varphi) \mapsto (\vartheta, \varphi + s)$ wherever that chart applies. The rotation itself is defined globally, including when a trajectory crosses the longitude cut. Its fixed points are the poles.

10.3 Lie differentiation

Fix p and let $q = \Phi_t(p)$. To compare Y_q with Y_p , use the inverse flow to bring Y_q back to T_pM ,

$$Y_q \in T_qM \quad \longmapsto \quad d(\Phi_{-t})_q(Y_q) \in T_pM.$$

This is a comparison supplied by the flow of X , not by a connection. The Lie derivative measures the resulting first-order change.

Definition 10.4: Lie derivative

Let Φ_t be the local flow of X . For a vector field Y ,

$$(\mathcal{L}_X Y)_p = \lim_{t \rightarrow 0} \frac{d(\Phi_{-t})_{\Phi_t(p)}(Y_{\Phi_t(p)}) - Y_p}{t}.$$

For a covariant tensor field A ,

$$\mathcal{L}_X A = \left. \frac{d}{dt} \right|_{t=0} \Phi_t^* A.$$

For a scalar field, $\mathcal{L}_X f = X(f)$. On mixed tensors, Lie differentiation is determined by the tensor product rule and compatibility with contraction.

In the vector formula, both terms in the numerator lie in T_pM . For covariant tensors, pullback gives the corresponding comparison at the same point. This explains the opposite directions of the two maps: vectors are brought back with the inverse differential, while covariant tensors are brought back by pullback.

Theorem 10.5: Lie derivative of a vector field

For smooth vector fields X, Y ,

$$\mathcal{L}_X Y = [X, Y].$$

The condition $[X, Y] = 0$ is equivalent to preservation of Y by the local flow of X , meaning

$$d(\Phi_t)_p(Y_p) = Y_{\Phi_t(p)}.$$

It is also equivalent to local commutation of the flows of X and Y .

Proof. In one chart, the flow and the differential of its inverse have the first-order expansions

$$\begin{aligned} \Phi_t^i(x) &= x^i + tX^i(x) + O(t^2), \\ (d\Phi_{-t})^i_j|_{\Phi_t(x)} &= \delta^i_j - t\partial_j X^i(x) + O(t^2). \end{aligned}$$

Also, $Y^i(\Phi_t(x)) = Y^i(x) + tX^j\partial_j Y^i(x) + O(t^2)$. Multiplying gives

$$d(\Phi_{-t})_{\Phi_t(p)}(Y_{\Phi_t(p)}) = Y_p + t(X^j\partial_j Y^i - Y^j\partial_j X^i)_p \partial_i|_p + O(t^2).$$

The defining limit is therefore (10.1). The term $X^j\partial_j Y^i$ comes from the change in Y along the flow. The term $-Y^j\partial_j X^i$ comes from bringing its value back by the inverse differential.

Write Φ_t^*Y for the vector field obtained by the same inverse-differential comparison. The flow law gives

$$\frac{d}{dt}\Phi_t^*Y = \Phi_t^*(\mathcal{L}_XY).$$

Thus $\mathcal{L}_XY = 0$ is equivalent to $\Phi_t^*Y = Y$, which is the stated preservation condition. When Φ_t preserves Y , it maps each integral curve of Y to an integral curve of Y . Uniqueness of integral curves makes the two flows commute locally. Conversely, differentiating their commutation in the flow parameter of Y gives the preservation condition. $\square$

Example 10.6: The order of two flows

On $\mathbb{R}^2$, let $X = \partial_x$ and $Y = x\partial_y$. Their flows are $\Phi_t(x, y) = (x+t, y)$ and $\Psi_s(x, y) = (x, y+sx)$. The two orders give

$$\Psi_s \circ \Phi_t(x, y) = (x+t, y+sx+st), \quad \Phi_t \circ \Psi_s(x, y) = (x+t, y+sx).$$

They differ in the y -direction by st . Equivalently,

$$\Psi_{-s} \circ \Phi_{-t} \circ \Psi_s \circ \Phi_t(x, y) = (x, y+st).$$

The residual displacement agrees with $[X, Y] = \partial_y$. The bracket detects the failure of the two flows to commute.

When frame fields have commuting flows, their flow parameters can serve as independent coordinates. This gives a criterion for recognizing coordinate frames.

Proposition 10.7: When a frame comes from coordinates

Smooth frame fields $e_1, \dots, e_n$ form a coordinate frame locally around each point if and only if $[e_i, e_j] = 0$ for all i, j .

Proof. Coordinate partial derivatives commute. Conversely, suppose the frame fields commute, with local flows Φ_i^t . Fix a point p and set

$$F(t^1, \dots, t^n) = \Phi_1^{t^1} \circ \dots \circ \Phi_n^{t^n}(p)$$

for sufficiently small parameters. The differential at zero sends the standard basis to $e_1|_p, \dots, e_n|_p$, so it is invertible. The inverse function theorem makes F a local parametrization. Commutation allows each flow factor to be differentiated independently, giving $dF(\partial/\partial t^i) = e_i$. Hence the inverse map defines coordinates with the required coordinate frame. $\square$

Preservation of Y does not mean that $Y(f)$ is constant along X for every function f . For instance, ∂_x and ∂_y commute, but $\partial_y(xy) = x$ varies along ∂_x . The bracket condition says that the two orders of differentiation agree, not that their common value is zero.

10.4 Lie derivatives of differential forms

For forms, the Lie derivative can be computed without solving for the flow explicitly. The required additional operation is insertion of the vector field into a form.

For a k -form α , $k \geq 1$, the *interior product* $\iota_X\alpha$ inserts X into the first argument,

$$(\iota_X\alpha)(X_2, \dots, X_k) = \alpha(X, X_2, \dots, X_k).$$

It is a $(k-1)$ -form. For a one-form, $\iota_X \alpha = \alpha(X)$ is a function. Set $\iota_X f = 0$ for functions. The wedge-product rule is

$$\iota_X(\alpha \wedge \beta) = \iota_X \alpha \wedge \beta + (-1)^k \alpha \wedge \iota_X \beta, \quad \alpha \in \Omega^k(M).$$

Theorem 10.8: Cartan's formula

For every differential form α ,

$$\mathcal{L}_X \alpha = \iota_X d\alpha + d(\iota_X \alpha). \quad (10.3)$$

Proof. Both sides preserve degree. On the right, one operation raises the degree and the other lowers it. Expanding their graded product rules shows that the mixed terms cancel, leaving the ordinary product rule for wedge products. On functions, the right-hand side is $\iota_X df = X(f)$. Naturality of d under pullback gives $\mathcal{L}_X df = d(X(f))$, which also agrees with the right-hand side. Locally, every form is a sum of products of functions and coordinate differentials. Agreement on these factors and the product rule therefore imply agreement on every form. $\square$

Lie differentiation of the scalar pairing gives

$$X(\alpha(Y)) = (\mathcal{L}_X \alpha)(Y) + \alpha([X, Y]).$$

For $\alpha = \alpha_i dx^i$, the product rule and $\mathcal{L}_X dx^i = d(X^i)$ give

$$(\mathcal{L}_X \alpha)_i = X^j \partial_j \alpha_i + \alpha_j \partial_i X^j. \quad (10.4)$$

The second term is needed because the flow also acts on the covector basis. Lie differentiation preserves form degree, while exterior differentiation raises it. The two operations commute, $\mathcal{L}_X d\alpha = d\mathcal{L}_X \alpha$.

10.5 Exterior differentiation in a general frame

The coordinate formula for $d\alpha$ uses commuting coordinate fields. For a one-form α and arbitrary vector fields X, Y , the corresponding formula includes a bracket term,

$$(d\alpha)(X, Y) = X(\alpha(Y)) - Y(\alpha(X)) - \alpha([X, Y]). \quad (10.5)$$

Expanding in coordinates shows why: derivatives of the components of X and Y cancel against the bracket term, leaving $(\partial_i \alpha_j - \partial_j \alpha_i) X^i Y^j$.

Let $\varepsilon^1, \dots, \varepsilon^n$ be dual to a local frame $e_1, \dots, e_n$, and define its commutator coefficients by $[e_b, e_c] = C^a_{bc} e_a$. Since $\varepsilon^a(e_b) = \delta^a_b$ is constant, the first two terms of (10.5) vanish, leaving $(d\varepsilon^a)(e_b, e_c) = -C^a_{bc}$. Expanding in the coframe gives

$$d\varepsilon^a = -\frac{1}{2} C^a_{bc} \varepsilon^b \wedge \varepsilon^c. \quad (10.6)$$

The functions C^a_{bc} depend on the chosen frame and do not transform as tensor components under an arbitrary varying frame change. For a coordinate coframe they vanish, giving $d(dx^i) = 0$. A general coframe need not consist of differentials of coordinates.

Lie differentiation and covariant differentiation

Lie differentiation compares a field using the flow of X . Covariant differentiation compares it using a chosen connection. These rules need not agree. For the flat connection on $\mathbb{R}^2$, take $X = x\partial_x + y\partial_y$ and $Y = \partial_x$. Then

$$\nabla_X Y = 0, \quad \mathcal{L}_X Y = [X, Y] = -Y.$$

The flow of X is a dilation and rescales tangent vectors. Its comparison differs from parallel transport for the flat connection, which preserves the constant Cartesian field. Moreover, $\mathcal{L}_X Y$ is not linear over $C^\infty(M)$ in X , whereas $\nabla_X Y$ must be.

Student work – rotation and the sphere one-form

For the sphere rotation field $K = \partial_\varphi$ and the one-form $\beta = \iota^*(x dy - y dx)$, compute $\iota_K \beta$ and $\iota_K d\beta$. Show that $\mathcal{L}_K \beta = 0$ using both Cartan's formula and the rotation flow.

11 Metrics and measurement

SOURCE SLIDE 18

A smooth manifold supplies tangent vectors and covectors, but it does not assign scalar products to tangent vectors. A metric supplies a bilinear form in each tangent space, with smooth dependence on the base point. This is the pointwise generalization of the Minkowski scalar product used in SR1.

11.1 Metrics and signature

Definition 11.1: Pseudo-Riemannian metric

A *pseudo-Riemannian metric* is a smooth tensor field g of type $(0, 2)$ whose value at each $p \in M$ is a symmetric, nondegenerate bilinear form

$$g_p : T_p M \times T_p M \longrightarrow \mathbb{R}.$$

Nondegeneracy means

$$[g_p(v, w) = 0 \text{ for every } w \in T_p M] \implies v = 0.$$

The pair (M, g) is a *pseudo-Riemannian manifold*.

Both input vectors belong to the same tangent space. In coordinates,

$$g = g_{ij} dx^i \otimes dx^j, \quad g_{ij} = g(\partial_i, \partial_j). \quad (11.1)$$

The component matrix is symmetric and invertible. Smoothness means that each coefficient g_{ij} is smooth. On a chart overlap,

$$g'_{ab} = \frac{\partial x^i}{\partial y^a} \frac{\partial x^j}{\partial y^b} g_{ij}.$$

The components change, but their evaluation on the same two vectors gives the same scalar.

Every nondegenerate real symmetric bilinear form has a basis in which its matrix is $\text{diag}(I_{n_+}, -I_{n_-})$, with $n_+ + n_- = n$. The numbers of positive and negative entries are independent of the basis. Their pair (n_+, n_-) is the *signature*. For a smooth nondegenerate metric, the signature is constant on each connected component. A change of sign of an eigenvalue would require it to pass through zero, contradicting nondegeneracy.

A positive-definite metric has signature $(n, 0)$ and is called *Riemannian*. A *Lorentzian* metric has one sign different from the others. In four dimensions we use $(n_+, n_-) = (1, 3)$, or $(+, -, -, -)$.

For a Riemannian metric, lengths and angles at a point are

$$\|v\|_g = \sqrt{g(v, v)}, \quad \cos \theta = \frac{g(v, w)}{\|v\|_g \|w\|_g} \quad (v, w \neq 0).$$

The length of a piecewise smooth curve is

$$L_g(\gamma) = \int_a^b \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt.$$

Under a regular reparametrization, the integrand acquires the factor $|dt/ds|$, so the integral is unchanged. On a connected Riemannian manifold, the distance between two points is the infimum of the lengths of curves joining them. It is not generally the length of an arbitrarily chosen joining curve.

Nondegeneracy and null vectors

A Lorentzian metric can have $g(v, v) = 0$ for $v \neq 0$. Nondegeneracy excludes a nonzero vector orthogonal to *every* vector, not one orthogonal to itself. Thus $\sqrt{|g(v, v)|}$ is not a norm, and the Riemannian angle formula does not apply to arbitrary Lorentzian vectors.

With our signature, a nonzero vector is timelike, null, or spacelike according as $g(v, v) > 0$, $g(v, v) = 0$, or $g(v, v) < 0$. Distinguishing future from past also requires a consistent choice of time orientation.

11.2 Metric duality and the inverse metric

Fix $v \in T_p M$ and allow the second argument of $g_p(v, w)$ to vary. The result is a covector, denoted by $v^\flat$,

$$v^\flat(w) = g_p(v, w), \quad \flat_p : T_p M \longrightarrow T_p^* M.$$

If $v^\flat = 0$, nondegeneracy gives $v = 0$. Since the two spaces have the same finite dimension, $\flat_p$ is an isomorphism. Its inverse is denoted by $\sharp_p : T_p^* M \rightarrow T_p M$. These are the *musical isomorphisms*.

Let (g^{ij}) be the inverse of (g_{ij}) , so that $g^{ik} g_{kj} = \delta^i_j$. Then

$$v^\flat = g_{ij} v^j dx^i, \quad \alpha^\sharp = g^{ij} \alpha_j \partial_i. \quad (11.2)$$

These are index lowering and raising. The maps vary smoothly with the point and therefore also act on fields. Unlike evaluation $\alpha(v)$, they depend on the metric.

The inverse metric is a tensor of type $(2, 0)$,

$$g^{-1} = g^{ij} \partial_i \otimes \partial_j, \quad g^{-1}(\alpha, \beta) = g(\alpha^\sharp, \beta^\sharp) = g^{ij} \alpha_i \beta_j.$$

It acts on covectors rather than vectors. Even when the matrices (g_{ij}) and (g^{ij}) have the same numerical entries, they represent tensors of different types.

The differential df gives the rate of change of f in any direction. The metric lets us express this same linear measurement as a scalar product with a vector. That vector field is the *metric gradient*, obtained by raising the index of df ,

$$\text{grad}_g f = (df)^\sharp = g^{ij} (\partial_j f) \partial_i, \quad g(\text{grad}_g f, X) = X(f). \quad (11.3)$$

The one-form df is defined by the smooth structure alone. The vector $\text{grad}_g f$ is defined only after choosing g .

12 Sphere and spacetime metrics

SOURCE SLIDE 19

12.1 The round metric on the sphere

The embedding $\iota : S^2 \hookrightarrow \mathbb{R}^3$ lets us measure tangent vectors using the Euclidean metric δ of the ambient space. Its pullback is the *induced metric*,

$$g_{S^2} = \iota^* \delta, \quad (g_{S^2})_p(v, w) = \delta_{\iota(p)}(d\iota_p v, d\iota_p w).$$

The differential $d\iota_p$ is injective, so a nonzero tangent vector has a nonzero ambient representative. Its Euclidean square is positive, making the induced metric Riemannian.

Using the coordinate vectors from Section 6,

$$g_{\vartheta\vartheta} = e_\vartheta \cdot e_\vartheta = 1, \quad g_{\vartheta\varphi} = 0, \quad g_{\varphi\varphi} = e_\varphi \cdot e_\varphi = \sin^2 \vartheta.$$

Thus

$$g_{S^2} = d\vartheta \otimes d\vartheta + \sin^2 \vartheta d\varphi \otimes d\varphi. \quad (12.1)$$

The factor $\sin^2 \vartheta$ reflects the smaller radius of a latitude circle, since $\|e_\varphi\| = \sin \vartheta$. The component order is (ϑ, φ) . Reversing that order gives $\text{diag}(\sin^2 \vartheta, 1)$, without changing the metric. For a sphere of radius R , the angular metric is multiplied by R^2 .

On this angular chart, the inverse metric gives

$$\text{grad} f = (\partial_\vartheta f) \partial_\vartheta + \csc^2 \vartheta (\partial_\varphi f) \partial_\varphi.$$

The formula does not apply at the poles because the angular chart does not include them. The apparent singularity of its coefficients does not imply that the metric or a smooth gradient is singular there.

Rescaling the coordinate vectors gives the orthonormal frame and its dual coframe,

$$E_1 = \partial_\vartheta, \quad E_2 = \csc \vartheta \partial_\varphi, \quad \varepsilon^1 = d\vartheta, \quad \varepsilon^2 = \sin \vartheta d\varphi.$$

In this frame,

$$g_{S^2} = \varepsilon^1 \otimes \varepsilon^1 + \varepsilon^2 \otimes \varepsilon^2.$$

The metric components are constant, but the frame is not a coordinate frame. Indeed,

$$[E_1, E_2] = -\cot \vartheta E_2, \quad d\varepsilon^1 = 0, \quad d\varepsilon^2 = \cot \vartheta \varepsilon^1 \wedge \varepsilon^2.$$

The bracket vanishes on the equator, but not on any nonempty open subset of the angular chart. By the coordinate-frame criterion, no coordinates on such a set have E_1, E_2 as their coordinate vectors. Constant metric components in a varying noncoordinate frame therefore do not establish flatness.

The rotation field and the one-form introduced earlier are now related,

$$K = \partial_\varphi, \quad K^\flat = \sin^2 \vartheta d\varphi = \beta, \quad g(K, K) = \sin^2 \vartheta.$$

Their definitions did not require a metric. The identification $K^\flat = \beta$ uses the particular round metric.

The stereographic chart is regular at the south pole. Differentiating its inverse parametrization gives

$$g_{S^2} = \frac{4}{(1+u^2+v^2)^2} (du \otimes du + dv \otimes dv). \quad (12.2)$$

At $(u, v) = (0, 0)$, the matrix is $4I_2$. This describes the same metric in coordinates that remain valid where the angular chart fails.

Student work – the sphere metric in two frames

Verify the orthonormal frame and coframe identities above, then derive (12.2).

Student work – the gradient of height

For $f = z$ on S^2 , compute df , $\text{grad } f$, and $g(\text{grad } f, \text{grad } f)$. Find the gradient at the poles using its ambient tangent-vector representation.

12.2 Lorentzian line elements and proper time

In inertial coordinates $x^\mu = (ct, x, y, z)$, the Minkowski metric is

$$\eta = \eta_{\mu\nu} dx^\mu \otimes dx^\nu, \quad (\eta_{\mu\nu}) = \text{diag}(1, -1, -1, -1).$$

A general spacetime metric is written $g = g_{\mu\nu}(x) dx^\mu \otimes dx^\nu$. Its *line element* is the quadratic-form notation

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu. \quad (12.3)$$

Along a curve $\gamma(\lambda)$, this means

$$ds^2 = g_{\gamma(\lambda)}(\dot{\gamma}, \dot{\gamma}) d\lambda^2, \quad \dot{\gamma} = \frac{d\gamma}{d\lambda}.$$

The products $dx^\mu dx^\nu$ in a line element are not wedge products. A symmetric tensor cannot be represented by $g_{\mu\nu} dx^\mu \wedge dx^\nu$, since that expression vanishes by antisymmetry.

In affine Minkowski coordinates, the finite displacement between two events has invariant square $\Delta s^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu$. On a general manifold, coordinate differences do not form the components of an intrinsic displacement vector. Therefore, $g_{\mu\nu}(p) \Delta x^\mu \Delta x^\nu$ is not an exact invariant separation of arbitrary events. The line element instead evaluates the metric on a tangent vector at one point of a curve.

For a future-directed timelike worldline in a time-oriented spacetime, the elapsed proper time is

$$\tau[\gamma] = \frac{1}{c} \int_a^b \sqrt{g_{\gamma(\lambda)}(\dot{\gamma}(\lambda), \dot{\gamma}(\lambda))} d\lambda. \quad (12.4)$$

Here ds^2 has units of length squared. The integral depends on the worldline, not only on its endpoints. With proper time as the parameter, the four-velocity $U = d\gamma/d\tau$ satisfies $g(U, U) = c^2$, or $g(U, U) = 1$ in units $c = 1$.

12.3 Local Lorentz frames

At any event p , the Lorentzian metric admits a basis $e_0|_p, \dots, e_3|_p$ with $g_p(e_a, e_b) = \eta_{ab}$. An orthonormal frame can also be chosen smoothly on a sufficiently small neighborhood. Extend the basis at p to local fields and orthonormalize using the metric. After shrinking the neighborhood, the normalization factors remain nonzero, so the construction is smooth.

Write the frame and its dual coframe as

$$e_a = e_a^\mu \partial_\mu, \quad \varepsilon^a = \varepsilon^a_\mu dx^\mu, \quad \varepsilon^a(e_b) = \delta^a_b.$$

Then

$$g = \eta_{ab} \varepsilon^a \otimes \varepsilon^b, \quad g_{\mu\nu} = \eta_{ab} \varepsilon^a_\mu \varepsilon^b_\nu.$$

Here $a, b = 0, 1, 2, 3$ label the orthonormal frame, while μ, ν label the coordinate basis. The frame need not be a coordinate frame, just as in the sphere example.

To retain the component convention of SR1, write another orthonormal frame as

$$e'_a = e_b (\Lambda^{-1})^b_a, \quad \Lambda^\top \eta \Lambda = \eta.$$

The components of the same vector then satisfy $v'^a = \Lambda^a_b v^b$. Thus Lorentz transformations relate orthonormal bases of a tangent space. A change of spacetime coordinates has a general invertible Jacobian and need not relate orthonormal coordinate bases. These are different uses of a basis transformation.

A linear change of a chart can make its coordinate basis agree with a prescribed basis at p . In particular, one can arrange $g_{\mu\nu}(p) = \eta_{\mu\nu}$. This fixes the metric components at one event. It does

not make them constant on a neighborhood or make their first derivatives vanish. Appendix A.6 constructs the stronger local inertial coordinate choice.

12.4 Metric invariance under a flow

A diffeomorphism $F : M \rightarrow M$ is an *isometry* when $F^*g = g$. Equivalently,

$$g_{F(p)}(dF_p v, dF_p w) = g_p(v, w)$$

for all p, v, w . The map sends points and vectors to new ones while preserving their metric pairings. This is an additional property of the map, not an automatic consequence of changing coordinates.

The Lie derivative tests whether a flow consists of local isometries. For the metric, the tensor product rule gives

$$(\mathcal{L}_X g)_{ij} = X^k \partial_k g_{ij} + g_{kj} \partial_i X^k + g_{ik} \partial_j X^k. \quad (12.5)$$

The first term measures how the metric coefficients vary along X . The other two account for the flow's action on the two vector arguments. The condition $\mathcal{L}_X g = 0$ is equivalent to $\Phi_t^* g = g$ on the flow domains. Such an X is called a *Killing vector field*. This definition requires a metric and a vector field, but no connection.

For the round sphere, $K = \partial_\varphi$ has constant components in the angular chart and the metric coefficients are independent of φ . Hence $\mathcal{L}_K g = 0$. Its flow is the rotation already constructed, which preserves the induced metric globally. Since $\beta = g(K, \cdot)$, the product rule also gives

$$\mathcal{L}_K \beta = (\mathcal{L}_K g)(K, \cdot) + g(\mathcal{L}_K K, \cdot) = 0.$$

This relates the earlier invariance of the sphere one-form to the symmetry of the metric.

12.5 From the metric to covariant geometry

SOURCE SLIDE 20

A smooth manifold has local charts and a tangent space at each point. Charts give components, while transition maps relate the descriptions of the same tensors and forms. Exterior differentiation uses only the smooth structure. Lie differentiation compares fields through a local flow, while a connection defines covariant differentiation and parallel transport along curves. A metric supplies a nondegenerate bilinear pairing within each tangent space.

To transport vectors along a curve while preserving their metric pairing, choose a metric-compatible connection. Requiring zero torsion as well selects the Levi-Civita connection. Appendix A derives this connection and completes the sphere transport calculation. Appendix B applies these constructions to a noninertial chart on Minkowski spacetime.

A The Levi–Civita connection

An affine connection and a metric are independent choices in general. For a given metric, requiring compatibility with its bilinear forms and requiring vanishing torsion selects a unique connection. To state these conditions precisely, first extend covariant differentiation from vectors to covectors and tensor fields.

A.1 Covectors and tensor fields

For a one-form α and a vector field Y , the pairing $\alpha(Y)$ is a scalar function. We require its derivative to obey

$$X(\alpha(Y)) = (\nabla_X \alpha)(Y) + \alpha(\nabla_X Y).$$

Solving for the first term defines covariant differentiation of covectors,

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

This is $C^\infty(M)$ -linear in Y , so it is a one-form. In coordinates,

$$\nabla_i \alpha_j := (\nabla_{\partial_i} \alpha)(\partial_j) = \partial_i \alpha_j - \Gamma^k_{ij} \alpha_k. \quad (\text{A.1})$$

The opposite signs of the connection terms ensure the ordinary product rule for the scalar contraction $\alpha_j Y^j$.

For general tensors, require the Leibniz rule for tensor products and compatibility with contractions. There is one positive connection term for each upper index and one negative term for each lower index. For example,

$$(\nabla_i T)^a{}_b = \partial_i T^a{}_b + \Gamma^a_{ic} T^c{}_b - \Gamma^c_{ib} T^a{}_c.$$

For a scalar field, $\nabla_i f = \partial_i f$. These first derivatives are the components of the one-form df . Differentiating once more therefore uses the covector rule, giving the *covariant Hessian*,

$$(\nabla df)_{ij} = \partial_i \partial_j f - \Gamma^k_{ij} \partial_k f. \quad (\text{A.2})$$

Unlike the uncorrected second partial derivatives, this is a covariant two-tensor. This tensor is also denoted by $\text{Hess}_\nabla f$. Its symmetry will depend on the torsion of the connection.

Student work – the difference of two connections

Prove that the difference of two affine connections defines a tensor of type $(1, 2)$. Verify the result from the transformation law of their connection coefficients.

A.2 Torsion and metric compatibility

The first condition relates the antisymmetric part of covariant differentiation to the Lie bracket. The second controls how scalar products change under differentiation.

Definition A.1: Torsion

The *torsion* of an affine connection ∇ is

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

The connection is *torsion free* if $T(X, Y) = 0$ for all vector fields X, Y .

The derivative terms from the Leibniz rules cancel, so T is $C^\infty(M)$ -bilinear and defines a tensor of type $(1, 2)$. In a coordinate frame, $[\partial_i, \partial_j] = 0$, and therefore

$$T^k_{ij} = \Gamma^k_{ij} - \Gamma^k_{ji}.$$

Consequently, torsion vanishes exactly when the lower indices of the connection coefficients are symmetric in every coordinate frame. For a general frame e_a , define $\nabla_{e_a} e_b = \Gamma^c_{ab} e_c$ and $[e_a, e_b] = C^c_{ab} e_c$. Then

$$T^c_{ab} = \Gamma^c_{ab} - \Gamma^c_{ba} - C^c_{ab}.$$

The bracket term must be included when the frame is not a coordinate frame.

Definition A.2: Metric-compatible connection

An affine connection is *compatible with g* if

$$X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

for all vector fields X, Y, Z . Equivalently, the covariant derivative of the metric vanishes, $\nabla g = 0$.

In coordinates, compatibility reads

$$\partial_i g_{jk} = \Gamma^\ell_{ij} g_{\ell k} + \Gamma^\ell_{ik} g_{j\ell}. \quad (\text{A.3})$$

This condition links differentiation of the metric components to the connection, rather than requiring the components to be constant.

A.3 Existence and uniqueness

Theorem A.3: Levi-Civita connection

Every pseudo-Riemannian metric g determines a unique torsion-free, metric-compatible affine connection. This is its *Levi-Civita connection*.

To determine $\nabla_X Y$, it is enough to determine its scalar product with every vector field Z . Nondegeneracy then fixes the vector itself. Combining metric compatibility with zero torsion gives the *Koszul formula*,

$$\begin{aligned} 2g(\nabla_X Y, Z) = & X(g(Y, Z)) + Y(g(Z, X)) - Z(g(X, Y)) \\ & - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]). \end{aligned} \quad (\text{A.4})$$

To derive it, expand the three differentiated scalar products using compatibility. Their signs are chosen so that the unwanted derivatives cancel after using $\nabla_X Y - \nabla_Y X = [X, Y]$, together with its cyclic versions. The remaining derivative is $2g(\nabla_X Y, Z)$, and the interchanges produce the three bracket terms displayed above.

The right-hand side depends only on g, X, Y, Z , so nondegeneracy gives uniqueness. For existence, use the same formula as a definition and check the required properties. Denote half of its right-hand side by $B(X, Y, Z)$. The bracket and product rules give $B(X, Y, fZ) = fB(X, Y, Z)$. In this calculation the terms containing $X(f)g(Y, Z)$ and $Y(f)g(X, Z)$ cancel in pairs. Hence B defines a smooth one-form in Z for fixed X, Y . Applying $\sharp$ to this one-form defines a unique smooth field $\nabla_X Y$.

It remains to check that this field defines a connection and has the two promised properties. The same rules give

$$B(fX, Y, Z) = fB(X, Y, Z), \quad B(X, fY, Z) = X(f)g(Y, Z) + fB(X, Y, Z).$$

Together with real bilinearity, these are the connection axioms. The sum and difference identities are

$$\begin{aligned} B(X, Y, Z) + B(X, Z, Y) &= X(g(Y, Z)), \\ B(X, Y, Z) - B(Y, X, Z) &= g([X, Y], Z). \end{aligned}$$

The first is metric compatibility. By nondegeneracy, the second gives $\nabla_X Y - \nabla_Y X = [X, Y]$. This proves existence as well as uniqueness, without first choosing coordinates.

Proposition A.4: Levi-Civita coefficients

In a coordinate frame, the Levi-Civita connection has coefficients

$$\Gamma^k_{ij} = \frac{1}{2}g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}). \quad (\text{A.5})$$

To see the coordinate calculation directly, write the three metric compatibility equations and use the symmetry of the lower connection indices. Adding the equations with differentiation directions i, j and subtracting the equation with direction ℓ gives

$$\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij} = 2g_{k\ell} \Gamma^k_{ij}.$$

Multiplication by $g^{m\ell}$ gives the stated formula. Equivalently, evaluate the Koszul formula on coordinate fields, whose brackets vanish. Formula (A.5) is specific to the Levi-Civita connection. A general affine connection is not determined by a metric.

Student work – constructing the metric connection

Derive the Koszul formula from metric compatibility and zero torsion. Verify the linearity and product-rule identities for B used in the existence proof.

A.4 Parallel transport and scalar products

For a metric-compatible connection and fields V, W along a curve,

$$\frac{d}{dt}g_{\gamma(t)}(V, W) = g_{\gamma(t)}\left(\frac{DV}{dt}, W\right) + g_{\gamma(t)}\left(V, \frac{DW}{dt}\right). \quad (\text{A.6})$$

The coordinate formula follows directly from (A.3), and is valid even when the fields are defined only along the curve. If both fields are parallel, the right-hand side vanishes. Thus parallel transport is a linear isometry between the endpoint tangent spaces,

$$g_{\gamma(b)}(P_{\gamma}v, P_{\gamma}w) = g_{\gamma(a)}(v, w).$$

It preserves Riemannian lengths and angles, and preserves the causal type of nonzero vectors in Lorentzian geometry. No positive-definite norm is needed for the latter statement.

Torsion also expresses the relation between the two derivatives of vector fields,

$$\mathcal{L}_X Y = [X, Y] = \nabla_X Y - \nabla_Y X - T(X, Y).$$

For the Levi–Civita connection, the torsion term is absent, but the term $\nabla_Y X$ remains. Lie invariance and parallelism are therefore still different conditions.

The Hessian in (A.2) is symmetric for a torsion-free connection, since the lower connection indices are symmetric. It need not vanish. By contrast, $d^2 f$ always vanishes. The covariant Hessian and the second exterior derivative are different constructions.

For a one-form α , combine (10.5) with the covector product rule to obtain

$$(d\alpha)(X, Y) = (\nabla_X \alpha)(Y) - (\nabla_Y \alpha)(X) + \alpha(T(X, Y)). \quad (\text{A.7})$$

Thus, for a torsion-free connection, $(d\alpha)_{ij} = \nabla_i \alpha_j - \nabla_j \alpha_i$. The symmetric connection coefficients cancel under antisymmetrization. This equality expresses an already intrinsic exterior derivative in terms of a connection. It does not make d dependent on that connection.

Student work – torsion and the Hessian

Show that for any affine connection,

$$\text{Hess}_\nabla f(X, Y) - \text{Hess}_\nabla f(Y, X) = -df(T(X, Y)).$$

Student work – metric symmetry

For the Levi-Civita connection, show that

$$(\mathcal{L}_X g)(Y, Z) = g(\nabla_Y X, Z) + g(Y, \nabla_Z X).$$

Conclude that a Killing field satisfies $\nabla_i X_j + \nabla_j X_i = 0$.

A.5 The sphere connection and transport around a latitude

For the round metric with line element $ds^2 = (d\vartheta)^2 + \sin^2 \vartheta (d\varphi)^2$, the only nonzero coordinate coefficients are

$$\Gamma^\vartheta_{\varphi\varphi} = -\sin \vartheta \cos \vartheta, \quad \Gamma^\varphi_{\vartheta\varphi} = \Gamma^\varphi_{\varphi\vartheta} = \cot \vartheta. \quad (\text{A.8})$$

For example, $\Gamma^\vartheta_{\varphi\varphi} = -\frac{1}{2}\partial_\vartheta(\sin^2 \vartheta)$, while $\Gamma^\varphi_{\vartheta\varphi} = \frac{1}{2}\csc^2 \vartheta \partial_\vartheta(\sin^2 \vartheta)$. All other terms vanish because $g_{\vartheta\vartheta} = 1$, $g_{\vartheta\varphi} = 0$, and the coefficients are independent of φ .

The same result follows without the Christoffel formula. For the embedded coordinate vectors, write the normal vector as σ . Direct differentiation gives

$$\partial_\vartheta e_\vartheta = -\sigma, \quad \partial_\vartheta e_\varphi = \partial_\varphi e_\vartheta = \cot \vartheta e_\varphi,$$

$$\partial_\varphi e_\varphi = -\sin \vartheta \cos \vartheta e_\vartheta - \sin^2 \vartheta \sigma.$$

Projecting away the normal terms gives (A.8). Tangential projection is metric compatible because normal and tangent vectors are orthogonal. It is torsion free because antisymmetrized ambient derivatives of tangent fields give their intrinsic Lie bracket. Uniqueness therefore identifies this projection connection with the Levi-Civita connection.

Now follow the latitude $\vartheta = \vartheta_0$, with $0 < \vartheta_0 < \pi$, using φ as the parameter. Along this circle, write a tangent field in the orthonormal frame as

$$V(\varphi) = a(\varphi)E_1 + b(\varphi)E_2, \quad E_1 = \partial_\vartheta, \quad E_2 = \csc \vartheta \partial_\varphi.$$

The connection coefficients give

$$\nabla_{\partial_\varphi} E_1 = \cos \vartheta_0 E_2, \quad \nabla_{\partial_\varphi} E_2 = -\cos \vartheta_0 E_1.$$

Consequently, the parallel-transport equation is

$$a' = \cos \vartheta_0 b, \quad b' = -\cos \vartheta_0 a.$$

These equations rotate the component pair at constant angular rate $-\cos \vartheta_0$. Writing $c_0 = \cos \vartheta_0$ and $\Delta\varphi = \varphi - \varphi_0$, their solution is

$$\begin{pmatrix} a(\varphi) \\ b(\varphi) \end{pmatrix} = \begin{pmatrix} \cos(c_0 \Delta\varphi) & \sin(c_0 \Delta\varphi) \\ -\sin(c_0 \Delta\varphi) & \cos(c_0 \Delta\varphi) \end{pmatrix} \begin{pmatrix} a(\varphi_0) \\ b(\varphi_0) \end{pmatrix}. \quad (\text{A.9})$$

The quantity $a^2 + b^2 = g(V, V)$ is preserved. The frame extends smoothly around the latitude even though a single longitude chart does not. After one circuit the frame returns to itself, but the vector has generally rotated by $-2\pi \cos \vartheta_0$, understood modulo 2π . Thus transport around a closed path need not be the identity.

This calculation also separates two meanings of transport. The coordinate fields commute, $[\partial_\varphi, \partial_\vartheta] = 0$, but

$$\nabla_{\partial_\varphi} \partial_\vartheta = \cot \vartheta \partial_\varphi.$$

Rotations preserve ∂_ϑ on the angular patch, but that field is not parallel along a general latitude. Lie invariance and parallelism are distinct even for a metric-compatible, torsion-free connection.

Student work – parallel transport on the sphere

Recover (A.9) using coordinate components. Determine the transport map after one circuit at $\vartheta_0 = \pi/2$ and $\vartheta_0 = \pi/3$, and verify preservation of the metric pairing.

A.6 Coordinates inertial at one event

The Levi–Civita connection permits a stronger coordinate choice than the pointwise orthonormalization in Section 12.3. First choose coordinates x with $x(p) = 0$ and $g_{\mu\nu}(p) = \eta_{\mu\nu}$. A quadratic coordinate change can leave this basis unchanged at p while adjusting the connection coefficients through its second derivatives. Define new coordinates locally through

$$x^\rho = y^\rho - \frac{1}{2} \Gamma^\rho_{\mu\nu}(p) y^\mu y^\nu.$$

The Jacobian at the origin is the identity, so the inverse function theorem makes this a valid local change of coordinates. Since the lower indices of the Levi–Civita coefficients are symmetric, their transformation law gives

$$\tilde{\Gamma}^\rho_{\mu\nu}(p) = \Gamma^\rho_{\mu\nu}(p) + \left. \frac{\partial^2 x^\rho}{\partial y^\mu \partial y^\nu} \right|_p = 0.$$

The metric at the point remains η , and metric compatibility then implies

$$\tilde{g}_{\mu\nu}(p) = \eta_{\mu\nu}, \quad \partial_\rho \tilde{g}_{\mu\nu}(p) = 0.$$

These are local inertial conditions at a single event. They do not require the second metric derivatives to vanish and do not establish flatness on a neighborhood. The analogous Riemannian construction replaces η by δ . The quadratic construction fixes only the coordinate data needed at the point, not a unique chart away from it.

B Boost coordinates in Minkowski spacetime

This application uses a noninertial coordinate chart on flat spacetime. It combines the metric, a Killing field, covariant differentiation, and proper time without introducing a gravitational field.

B.1 The chart and its metric

Set $c = 1$ and use inertial coordinates (t, x, y, z) . On the right-hand wedge $x > |t|$, introduce the rapidity coordinate χ and radial coordinate $\rho > 0$. Boosts preserve the hyperbolas $x^2 - t^2 = \rho^2$, and χ parametrizes motion along each one. Thus set

$$t = \rho \sinh \chi, \quad x = \rho \cosh \chi. \quad (\text{B.1})$$

The inverse is

$$\rho = \sqrt{x^2 - t^2}, \quad \chi = \text{artanh}(t/x).$$

The Jacobian determinant of $(\chi, \rho) \mapsto (t, x)$ is ρ , so it is nonzero on this domain. Differentiation gives

$$dt = \rho \cosh \chi d\chi + \sinh \chi d\rho, \quad dx = \rho \sinh \chi d\chi + \cosh \chi d\rho.$$

Substituting in the Minkowski line element cancels the mixed terms,

$$ds^2 = \rho^2(d\chi)^2 - (d\rho)^2 - (dy)^2 - (dz)^2. \quad (\text{B.2})$$

Thus, in the order (χ, ρ, y, z) ,

$$(g_{\mu\nu}) = \text{diag}(\rho^2, -1, -1, -1), \quad (g^{\mu\nu}) = \text{diag}(\rho^{-2}, -1, -1, -1).$$

The components vary, but the spacetime is still Minkowski spacetime because this metric was obtained by an invertible change of chart. The chart does not cover the wedge boundary. The inertial metric remains regular there.

B.2 Boost symmetry and covariant differentiation

The coordinate field

$$K = \partial_\chi = x\partial_t + t\partial_x$$

generates active Lorentz boosts. Its flow is $\chi \mapsto \chi + s$, with ρ, y, z fixed. In the passive boost convention of SR1, the same matrix is written $L(-s)$. As a global field on Minkowski spacetime,

$$g(K, K) = x^2 - t^2.$$

It is timelike on the left and right wedges, null on their boundaries when nonzero, and spacelike where $|t| > |x|$. It vanishes when $t = x = 0$. On the right wedge it is future directed for the standard time orientation.

The metric components in (B.2) do not depend on χ , and the components of $K = \partial_\chi$ are constant there. The Lie-derivative formula therefore gives $\mathcal{L}_K g = 0$. The nonzero Levi-Civita coefficients are

$$\Gamma^\rho_{\chi\chi} = \rho, \quad \Gamma^\chi_{\chi\rho} = \Gamma^\chi_{\rho\chi} = \frac{1}{\rho}. \quad (\text{B.3})$$

For example, $\Gamma^\rho_{\chi\chi} = \frac{1}{2}(-1)(-\partial_\rho \rho^2) = \rho$. Consequently,

$$[\partial_\chi, \partial_\rho] = 0, \quad \nabla_{\partial_\chi} \partial_\rho = \frac{1}{\rho} \partial_\chi.$$

The coordinate frame commutes, but it is not a parallel frame.

B.3 An accelerated clock

Consider a future-directed worldline with $\rho = \rho_0 > 0$ and constant y, z . Its proper time satisfies $d\tau = \rho_0 d\chi$, so

$$U = \frac{d\gamma}{d\tau} = \frac{1}{\rho_0} \partial_\chi, \quad g(U, U) = 1.$$

The four-acceleration is the covariant derivative of the four-velocity,

$$a = \nabla_U U = \frac{1}{\rho_0^2} \nabla_{\partial_\chi} \partial_\chi = \frac{1}{\rho_0} \partial_\rho.$$

It has invariant square

$$g(a, a) = -\frac{1}{\rho_0^2}, \quad g(a, U) = 0.$$

The orthogonality also follows by differentiating the constant normalization,

$$0 = U(g(U, U)) = 2g(\nabla_U U, U) = 2g(a, U).$$

The observer has constant spatial coordinate labels, but nonzero four-acceleration. Being at rest in this coordinate system is therefore not the same as moving inertially.

In inertial coordinates, take $\chi = \tau/\rho_0$ after choosing the origins of both parameters. The component arrays in the inertial coordinate basis are

$$(U^\mu) = (\cosh \chi, \sinh \chi, 0, 0), \quad (a^\mu) = \frac{1}{\rho_0} (\sinh \chi, \cosh \chi, 0, 0).$$

Ordinary proper-time differentiation is valid in this Cartesian inertial frame because its connection coefficients vanish. It gives the same acceleration and invariants. With length-valued ρ_0 and c restored,

$$d\tau = \frac{\rho_0}{c} d\chi, \quad \sqrt{-g(a, a)} = \frac{c^2}{\rho_0},$$

where $U = d\gamma/d\tau$ now has $g(U, U) = c^2$.

Student work – parallel transport along the accelerated worldline

Along the fixed- ρ_0 worldline, find the parallel field $V = A(\chi)\partial_\chi + B(\chi)\partial_\rho$ with $A(0) = A_0$, $B(0) = B_0$, in units $c = 1$. Verify that $\rho_0^2 A^2 - B^2$ is constant and explain why the observer's four-velocity is not parallel along the same curve.